



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Power Electronics for the Galway Energy Efficient Car 2025

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Abstract

The Galway Energy Efficient Car (GEEC) is a project developed by an interdisciplinary team of students from the University of Galway to compete in the annual Shell Eco-Marathon (SEM) competition in Europe. This project focuses on analysing the feasibility and advantages of upgrading the car's motor and improving its overall energy efficiency by refining the power electronics and optimising the driving strategy.

Building on the work of previous students work and thesis, this year's focus was on analysing the improvements of installing a more efficient motor system to improve the car's performance while reducing energy consumption. Careful comparison and selection of a new motor aim to achieve higher efficiency while under competition conditions, by improving both the electrical and mechanical systems of the vehicle.

Alongside the motor upgrade, significant work was undertaken to analyse and optimise the driving strategy used by the team. By studying factors such as speed profiles, acceleration patterns, and energy usage throughout a typical run in practice and during races, the project aims to develop a driving approach that minimises energy consumption and maximises efficiency during competition.

The insights gained from the theoretical motor improvements and strategic analysis will help the GEEC team make more informed decisions on vehicle setup and race tactics this year and in the future. These upgrades and strategies will play a critical role when the GEEC competes at the Shell Eco-Marathon in June 2025, which will be held in Silesia, Poland.

Overall, this project aims to contribute to the continued evolution of the GEEC, helping the team to push the boundaries of energy-efficient vehicle design and performance.

Declaration of Originality

I declare that this thesis is my original work except where stated.

Date:

Signature:

11/05/2025

Rory Egan

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There are many people I'd like to thank for their support throughout this project and all their help and advice was invaluable to me.

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η Efficiency (%)

ω Rotational Speed (Radians per second)

n Rotational Speed (Revolutions per minute)

k_M Motor mechanical torque constant (Nm/A)

k_n Motor mechanical speed constant (rpm/V)

R Motor terminal resistance (Ω)

V Motor terminal voltage (V)

I Motor terminal current (A)

I_0 Motor no load current (A)

List Of Acronyms

BLDC: Brushless Direct Current Motor

DAQ: Data Acquisition

ESC: Electronic Speed Controller

PWM: Pulse Width Modulation

RPM: Revolutions Per Minute

LiPo: Lithium Polymer Battery

IC: Integrated Circuit

Geec: Galway Energy Efficient Car

Li-ion: Lithium-ion Battery

BMS: Battery Monitoring System

SPICE: Simulation Program with Integrated Circuit Emphasis

EMF: Electromotive Force

PCB: Printed Circuit Board

SEM: Shell Eco-marathon

1 Introduction

The Galway Energy Efficient Car (GEEC) is the University of Galway's long-running entry into the Shell Eco-Marathon which is an international competition where student teams aim to design, build, and race the world's most energy-efficient vehicles. Our car first took to the track in 2015 where students have aimed to enhance different aspects of it each year.

This project continues that progress by analysing whether upgrading the GEEC's motor system from a traditional 24V brushed DC configuration to a modern 48V setup would be advantageous. This report details considering both brushed and brushless alternatives that can provide meaningful gains in performance and efficiency. The concept of this project is based on a well established principle of by lowering the current at the same power level by raising the system voltage, this can greatly minimise heat generation and resistive losses in cables, connectors, and drive electronics. Further advancements could be possible through decreased friction, increased dependability, and more accurate control when combined with brushless motor technology.

This project involves simulations, dyno and track testing. Using software tools like LTspice and PSIM, the performance of various motor types and configurations under different voltages was modelled to predict power losses, ripple currents, and efficiency trends. These models were then validated and expanded on through dynamometer and track testing, allowing real world comparisons and results between the current 24V system and proposed 48V alternatives. In addition to this, track testing with the existing vehicle was completed to study the impact of motor characteristics and driving trends on driving strategy especially in the context of the pulse and coasting techniques commonly used in the Shell Eco-Marathon to maximise range.

In essence, this work explores whether a shift to 48V could enable the GEEC to travel further using less energy, and whether it provides additional flexibility in fine tuning and changing how the car is driven during competition. The findings will not only inform decisions about future drivetrain upgrades but will also hopefully feed into the team's overall race planning and vehicle development in the years ahead.

1.1 The Shell Eco Marathon

The Shell Eco-Marathon [1] is a global student competition that's all about designing, building and testing the most efficient vehicle. Unlike normal and typical races where speed is king and the only goal, this one is different. The winner is the car that travels the farthest using the least amount of energy. Every year, university and secondary school teams from around the world take part in this challenge, building and racing vehicles that aim to squeeze out every last kilometre from their fuel or battery pack.

The event is split into three main regional competitions, one in the Americas, the Asia-Pacific and the Middle East, and Europe and Africa. Each region hosts its own version of the competition, and the best-performing teams may go on to compete in a global event after that. Last year in 2024, our team from the University of Galway travelled to the Circuit Paul Armagnac in Nogaro, France, to compete in the Europe and Africa round and this year, the Shell Eco-Marathon Europe and Africa event will take place at the Silesia Ring in Poland, with the latest version of the GEEC which was refined and tested during the year.

There are two classes of vehicles in the competition, Urban Concept and Prototype. The Urban Concept cars are closer to something you might normally see on a public road, with things like headlights, wind-wipers, and a focus on practicality to be used in the real world. The Prototype class which is where the GEEC competes, is all about innovation and pushing the limits of the cars each year to get the most efficient car. These vehicles are simpler and stripped out, aerodynamic, and designed purely for maximum energy efficiency.

On top of that, teams can choose from different energy sources with internal combustion engines, hydrogen fuel cells, or battery-electric systems all being options. The GEEC runs in the Prototype Battery Electric category, meaning our performance is measured in kilometres per kilowatt-hour (km/kWh). A joulemeter is installed by the organisers and records exactly how much energy we use during each timed attempt, and we only get four runs to post a score so every attempt and decision towards the cars setup during the race counts.

To succeed and win, all the teams need to do more than just build a light, streamlined and efficient vehicle. Driving strategy is just as important. One common tactic is known as “pulse and coast”, which is used by our team also, where the car gently accelerates to a target speed and then the motor is turned off to let it roll as far as possible before powering up again. It’s a delicate balance between conserving energy and staying above the minimum average speed required to log a valid run which is usually around 25 km/h. It's these kinds of strategic and engineering decisions that make the Eco-Marathon such a unique and demanding event each year.

1.2 The Galway Energy Efficient Car

The Galway Energy Efficient Car, or GEEC[2], is the University of Galway’s entry into the Shell Eco-Marathon. It's a student-led project that’s been running since 2015, with new students taking it on each year to improve on the previous car. The car competes in the Prototype Battery Electric category, where the aim is to travel as far as possible on as little energy as possible. The GEEC runs on a lithium polymer battery (usually a 6- or 7-cell pack) and up until now, has used a brushed DC motor for propulsion.



Figure 1-1 The GEEC Before A Race Attempt [2]

Over the years, the car has gone through lots of upgrades and modifications. Its best result so far was back in 2018 when it achieved 354 km/kWh, which is an incredible level of efficiency. In 2023, the team introduced a new carbon fibre monocoque chassis, helping to cut weight significantly. That year's score was 299 km/kWh which was not quite the record, but still impressive. For comparison, the winning car in last year's European competition managed over 1000 km/kWh. While the monocoque helped reduce mass, the car did face a few reliability problems that likely affected the final result along with possible weather and track conditions during the race.

The GEEC project is broken into several sub-teams like power electronics, drivetrain, chassis, windows and sealing, data acquisition, and race strategy. Each team is made up of a mix of students across all years, with final year students usually leading the sections. It's a really collaborative effort, and part of the idea is to pass down knowledge through the years so that the car and the team keep improving. Everyone involved is working toward the same main goal which is to make our car as efficient as possible and build on past projects.

1.3 Previous Iterations

Over the past number of years, various thesis projects have contributed significantly to improving the GEEC's power electronics, with a particular focus on the motor controller, wiring efficiency, and the electrical system as a whole.

1.3.1 Sorcha Tarpey (2016)

Sorcha contributed to the foundational stages of GEEC 2.0's electrical system. Her work involved optimising the entire power electronics system, from battery selection to motor suitability. Using PSIM, she analysed and modelled motor behaviour under various loads and thermal conditions. Her design adjustments aimed to improve system reliability and thermal performance. Although testing time was limited before the SEM event, her simulation results informed future design directions, particularly regarding overcurrent protection and motor selection strategies for future versions of the GEEC.

1.3.2 Tomás Crotty (2018)

Tomás developed and tested a high-efficiency motor drive circuit tailored for the GEEC's needs. His project placed emphasis on the controller hardware that interfaces with the brushed DC motor. Through hardware prototyping and simulation, he refined the PWM control and transistor switching methods. His work was crucial in establishing a reliable baseline design that would later be improved upon by students such as Anderson. Key takeaways from Tomás's work included techniques for noise suppression, thermal management, and component durability under race conditions.

1.3.3 Oisín Anderson (2020)

Oisín's project focused on developing a more efficient and robust motor controller for the GEEC using the foundation of the previous projects above. His work included a full review of the existing motor drive circuitry and wiring harness, identifying inefficiencies in the previous design, particularly in the existing system, which was only around 60% efficient. He designed and simulated a new wide-input motor controller, achieving simulated efficiencies of over 95%. A new wiring harness was also introduced, reducing energy losses to just 3% of what they had been previously. Though testing was constrained by external factors like covid, the theoretical performance showed significant promise for future use in competition settings.

1.4 Project Motivation

This project appealed to me because it offered a practical challenge in a real-world setting, with a clear goal of improving the energy efficiency of the GEEC through better motor and power system design. Working on a small part of a larger vehicle system, alongside a team all working toward the same goal, felt like a good opportunity to apply what I've learned during my degree in a hands-on, meaningful way while also being a part of a team.

My focus on the power electronics side of the car allowed me to explore potential improvements to the current 24V system, particularly by analysing whether moving to a 48V motor setup could offer better efficiency for the vehicle. Since all the power used to drive the car flows through this system, any gain here can make a real difference in competition and our overall score.

This project has been as much about learning and collaboration as it has been about technical performance. Helping to shape one small part of a larger system, and seeing that work integrated into a car representing our college, has made the whole process both rewarding and motivating.

2 Background Research and Current System

2.1 Motor Controllers

A motor controller is an electronic device that controls the operation of an electric motor by controlling how much power it receives, that then adjusts parameters like speed, torque, and direction. Its main function is to act as a link and brain between the power supply (such as a battery or power cell) and the motor, ensuring that power is delivered in a controlled and efficient manner when needed. There are several types of motor controllers, depending on the motor type and application. For brushed DC motors, controllers are generally quite simple and work by varying the voltage supplied to the motor using techniques like pulse-width modulation (PWM). For brushless DC (BLDC) motors, the controller is more complex and it must switch the current between the motor's phases in precise timing based on rotor position, essentially replacing the mechanical commutator found in brushed motors. Other motor controller types include those for AC motors, stepper motors, and servo motors, each with their own control strategies but will not be studied in this document. In electric vehicle applications like the GEEC, the motor controller is a critical part of the drivetrain, directly affecting the vehicle's energy efficiency and overall performance.

2.1.1 Brushed DC Motor Controller

A brushed DC motor controller is an electronic device that regulates the power delivered to a DC motor to control its speed, torque and direction. In a brushed motor, speed is directly proportional to the voltage applied to its terminals for a given load which is shown in the following equation. [3]

Equation 1 Speed Equation of Brushed DC Motor

$$\omega = \frac{V - RI}{K_e}$$

With this, most brushed motor controllers use a form of pulse-width modulation (PWM) to effectively adjust the average voltage seen by the motor. Basically, the controller rapidly switches the supply on and off (using power transistors like MOSFETs) to vary the duty cycle of the voltage, which in turn controls the motor's speed. The higher the duty cycle (on-time), the faster the motor will spin, assuming the load remains the same. An electronic control unit (for example, an Arduino or a dedicated driver IC) reads the throttle input from the driver and outputs a PWM signal to drive the power MOSFETs, which controls the motor's supply voltage. This method is efficient because the MOSFETs act as switches that spend most of their time either fully on or fully off that minimizes the time spent in low efficiency states.

2.1.2 Current Motor Controller

To get to the current motor controller in the GEEC first what it evolved from must be studied. Firstly it was a custom built brushed DC motor driver that functioned as a non-synchronous buck converter which stepped down the 24 V battery to the appropriate motor voltage. In this design, a single high-side MOSFET switched the DC supply to the motor, and a flyback diode provided a return path for current when the MOSFET switches itself off (allowing the motor's inductance to smoothly discharge). This controller was one of the first designs made for the car and is used presently mainly because it is a proven, working solution.

However, it had several practical and efficiency issues. Physically, it was built with multiple screw terminal connections between the battery, the DC/DC converter, and the Arduino control board. This construction was not ideal and during testing, it was common for wires to vibrate loose, causing short disconnections. The terminals were not well-labelled either, making maintenance tedious since it was difficult to trace which wire went where. Another issue was the lack of a proper pre-charge circuitry where the controller had large input filter capacitors that drew a big inrush current when the system is connected initially, the discharged capacitors appeared almost like a short circuit to the battery, causing a sudden surge of current when the controller is powered on. This surge was large enough to trigger the Battery Management System's over-current protection (OCP), preventing the car from starting until the system was reset.

The solution proposed for this is to incorporate a soft-start or pre-charge circuit so that the capacitors charge more slowly and avoid tripping the BMS. From an efficiency standpoint, the old GEEC controller also has limitations inherent to its non-synchronous buck design. When the switching MOSFET was off, current freewheels through a diode, which had a forward voltage drop. This causes some power loss as heat in the diode, making the conversion less efficient. At lower motor speeds (lower PWM duty cycles), the MOSFET is off for longer portions of time, meaning the diode conducts more often and these losses become significant. Oisín Anderson noted that the previous motor controller (which used this flyback diode design) could be as low as ~60% efficient in certain conditions. This means a large fraction of input power was wasted as heat, which is obviously detrimental in an efficiency-focused vehicle.

To address this, we moved to a synchronous buck topology, where the diode is replaced by a second MOSFET that switches in tandem (out of phase) with the first. In a synchronous controller, when the main transistor turns off, a second transistor turns on to provide a low resistance path for the inductive current. This basically fully eliminates the diode forward voltage drop losses, especially at low duty cycles, greatly improving efficiency. Anderson's thesis work involved designing a new motor controller with this synchronous design approach, achieving over 95% simulated efficiency under most operating conditions which was a huge improvement over the old ~60% efficient controller.

The trade off is increased complexity where the synchronous design requires careful timing (dead time to prevent both MOSFETs conducting simultaneously, which would cause a short) and often a dedicated gate driver IC to control the high-side transistor. Anderson implemented a half-bridge driver (the LM5104) to manage this, which simplified the Arduino control code. The GEEC's current controller is unidirectional as it only needs to drive the car forward so a single buck converter works and is used for speed control. In other applications where bidirectional control or regenerative braking is needed, an H-bridge (with four transistors) might be used to allow current to flow to or from the motor in both directions. For the GEEC, regenerative braking hasn't been a focus, so the controller's primary job is to efficiently throttle the motor's power while protecting the battery and motor from damage.

2.1.3 Brushless DC Motor Controller

Brushless DC (BLDC) motor controllers are more complicated than their brushed counterparts, mainly because BLDC motors don't have mechanical commutation. Instead of using brushes and a commutator to switch current through the windings, the controller does this job electronically using other components. It continuously switches the current between the motor's three phases in the correct sequence to keep the rotor spinning smoothly. This means a BLDC controller basically acts like a three-phase inverter, typically using six power transistors (usually MOSFETs) arranged in a three-phase bridge configuration.[4]

To do this properly and without faults, the controller needs to know the exact position of the rotor at all times. There are two main ways this can be done which are sensed and sensorless control. In sensed controllers, small Hall effect sensors are placed inside the motor and detect the rotor's magnetic field, which gives us the position. The controller reads these signals to determine which phases to energise to control the motor. In sensorless controllers, the controller estimates rotor position by monitoring the back-EMF generated in the unpowered phase and this makes the controller cheaper and more compact, but it can be less reliable at very low speeds or startup.[5]

The most common control method for BLDC motors is called trapezoidal control or six step commutation[6]. This method energises two of the three windings at a time and switches every 60 electrical degrees. It's a relatively simple and efficient concept, especially for motors operating at a constant speed. More advanced methods for the controller like field-oriented control (FOC) or sinusoidal control offer smoother torque and lower noise but are more complex and require a lot more design, testing and headaches.

One of the key benefits of a brushless controller is that it gives the user much more control over motors behaviour. Features can be added and customised like speed limiting, torque control, regenerative

braking, and temperature monitoring can all be implemented with this controller. The controller can also protect the motor by setting limits for current, voltage, and temperature, like a battery management system making the system more robust overall.

For the GEEC, switching to a BLDC motor would require designing or sourcing a controller that can handle the 48 V system, matching the selected motor's characteristics, and comply with Shell Eco-Marathon rules (i.e. be custom-built or documented). While it would add complexity compared to the brushed controller, the potential gains in efficiency, reliability, and control flexibility make it a strong candidate for future iterations of the car.

2.1.4 Brushed vs Brushless Motor Controllers

A brushed DC motor, as described briefly earlier, only requires a single output voltage to control its speed, since the motor's internal commutator handles all the phase switching automatically. In contrast, a brushless DC motor controller often referred to as an Electronic Speed Controller (ESC) is significantly more complex. BLDC motors typically have three separate windings (phases) that must be energised in a specific sequence to create a rotating magnetic field. The controller acts as a three-phase inverter, switching current between these coils in precise timing to rotate the motor. This switching is handled electronically using multiple MOSFETs arranged in a three-phase bridge layout. To do this properly, the controller needs to know the rotor's position which it gets from either built-in Hall effect sensors or by monitoring back-EMF in sensorless designs. A microcontroller or dedicated driver chip then generates the correct signals to switch the MOSFETs in sequence, adjusting the timing as the motor speeds up to keep rotation smooth and efficient. Unlike a brushed motor, where commutation is handled mechanically, the brushless controller performs all of this electronically, allowing for much more control over speed, torque, and even regenerative braking in some setups.

Despite the added complexity, moving to a brushless controller (and motor) can offer several key advantages. Firstly, brushless motors are typically more efficient and require less maintenance. With no brushes to wear down or cause friction, the main losses in the motor are from winding resistance and core magnetic losses. Many BLDC motors can achieve over 90% efficiency in their optimal range sometimes even exceeding the GEEC's current brushed motor, which peaks around 94%. Eliminating brushes also removes a common point of failure and the need for periodic servicing.

Another major advantage is the improved power-to-weight ratio. Since brushless motors don't have a commutator or brushes, they can be made smaller and lighter for the same power output. Most BLDC designs also place the windings on the stationary part of the motor (the stator), allowing for better cooling and simplified construction. For instance, a 200 W brushed motor might weigh around 1.1 kg, while a brushless equivalent could come in at roughly 0.3 kg. That weight saving can be significant in a competition like SEM, where every gram counts.[7]

Brushless motors also avoid issues with brush timing, especially at high speeds. In brushed motors as speed increases poor contacts between the brushes and commutator can limit the motors performance. A brushless controller, can dynamically adjust the timing through software to keep the motor running smoother across a wide speed range. This flexibility can be particularly useful for applications that demand better control or varying load conditions like throughout a race.

Regenerative braking is another area where brushless controllers have an edge. With the right setup, the ESC can act as a generator controller allowing the motor to feed energy back into the battery during deceleration. While not currently implemented in the GEEC due to having a brushed motor, this could be a useful addition for future designs. Brushed motors can also be used for regen with an H-bridge configuration, but it's generally more straightforward to implement with BLDC systems, particularly in commercial EVs where it's now standard.

Of course, there are trade-offs to switching to a brushless controller. The control circuitry is far more advanced and would need to be designed or carefully chosen to comply with Shell Eco-Marathon regulations, which require all motor controllers to be custom-developed for the vehicle. Electromagnetic interference (EMI) becomes more of a concern, and the integration of sensors (or the choice to go sensorless) must be carefully considered too. The motor itself would also have to be replaced, which might involve mechanical changes to the car such as mounting and drivetrain alignment if a motor of a different size was chosen.

2.2 Batteries

To improve the energy efficiency and performance of the GEEC, it is crucial to evaluate the design and configuration of its each part of the electrical powertrain. One key area of focus is the vehicle's battery system, which directly impacts power delivery, component sizing, and energy losses. Currently, the GEEC runs on a 24 V lithium-ion battery setup, but exploring a potential upgrade to a 48 V system. This shift could yield significant advantages in terms of electrical efficiency, wiring optimization, and thermal performance. The following sections outline the current system configuration, the technical details of both 24 V and 48 V systems, and a comparison of their respective benefits and trade-offs.

2.2.1 Current 24V Battery Setup

The GEEC is currently powered by a 24 V lithium-polymer (LiPo) battery pack composed of seven lithium-ion cells wired in series. Each cell has a nominal voltage of approximately 3.7 V, resulting in a total nominal pack voltage of 25.9 V. This is commonly referred to as a "24 V" battery because the voltage drops under load a phenomenon known as voltage sag [8] which brings the voltage closer to 24 V during operation.

LiPo batteries were selected over traditional Li-ion alternatives due to their superior peak discharge current characteristics and slightly higher nominal charge voltage. While LiPo batteries have lower energy density (175–225 Wh/L) compared to typical Li-ion cells (210–350 Wh/L)[9], this trade-off was ok for the GEEC where the total energy demand is relatively low and physical space is less of a constraint in the car. The battery is housed in an aluminium casing to protect it from impacts and to contain any fire in the rare event of a thermal runaway during the race. Safety is a critical concern with high-energy batteries, and because of this the pack is equipped with a Battery Management System (BMS).

The BMS monitors each cell's voltage, temperature, and the current drawn from the pack. It serves as a protective layer to prevent overcharge, over-discharge, and over current events where any of these could damage the battery or trigger dangerous conditions such as fire. According to Shell Eco-Marathon regulations, all lithium-based batteries must have a BMS capable of isolating the battery if an unsafe condition is detected[10]. In earlier versions of the GEEC, the BMS would cut off power during motor controller inrush or stall events when the current spiked to levels as high as 232 A. To avoid and stop this, the current system design includes features such as a soft-start designed by previous students.

In the current configuration, the 24 V battery supplies power to both the motor and the vehicle's electronic systems, with the them powered via a DC/DC converter. As energy is drawn during a run the voltage gradually drops, which can negatively affect performance. Lower voltage reduces the motor's ability to draw current at the same controller setting, thereby reducing power output especially towards the end of a race.

To combat this voltage sag, the GEEC uses a high-capacity 10 Ah battery pack. This helps maintain voltage stability and ensures the motor continues to receive sufficient power throughout a run. However even with a well chosen battery, the effects of voltage drop remain a challenge in longer runs or high-load scenarios.

LiPo batteries also come with specifications critical for performance tuning, including nominal voltage, maximum/minimum voltage, capacity (Ah), and charge/discharge ratings. The maximum, nominal, and minimum voltages for LiPo cells are typically 4.2 V, 3.7 V, and 3.3 V respectively. Battery capacity, expressed in ampere-hours (A*h) determines how long the battery can supply current before its fully discharged. The power capacity in watt-hours (W*h) can be calculated by multiplying the nominal voltage by the capacity in A*h.

$$Ah * C = \text{Maximum Current Outut}$$

Equation 2 Maximum Current Output For a Battery

Another key spec is the “C” rating, which defines how quickly a battery can charge or discharge relative to its capacity. For example, a 2 Ah battery with a 70C/40C rating can deliver 140 A peak and 80 A continuous. LiPo batteries typically have much higher C ratings than Li-ion batteries, making them better suited for high-current applications like electric vehicle propulsion, which is why it’s used in the GEEC. For instance, Turnigy [11] offers a 22 V, 5 Ah LiPo battery rated for 80C peak and 40C continuous discharge which far exceeds the 1–3C typical of standard Li-ion cells. This higher current capability reduces the need to wire multiple cells in parallel and simplifies the battery design and improves reliability.

Overall, the current 24 V battery setup balances power delivery, weight, safety, and reliability while also leaving some room for optimization especially in terms of efficiency and voltage stability over a full race cycle.

2.2.2 48V Battery Setup

This project has been evaluating a move to a 48 V system the maximum allowed by Shell Eco-Marathon rules. This transition would involve increasing the number of lithium-ion cells from 7 to 13 in series. A 13S lithium-ion pack, potentially with a lower capacity could still deliver equivalent or improved energy storage without significantly increasing the overall battery weight by too much.

Changing to a 48 V system would have several implications for the GEEC's powertrain. Firstly, all electrical components including the motor controller, connectors, and wiring must be rated to handle the full charge voltage of a 48 V pack, which is approximately 54.6 V. The current motor controller for a brushed dc motor was designed with this in mind and most of its components and wiring is rated for this, if a brushed dc motor was chosen.

One of the most significant advantages of moving to 48 V is the reduction in current draw for the same power output. Power is the product of voltage and current ($P = V \times I$), so doubling the voltage allows the same power to be delivered at half the current. This greatly reduces resistive (I^2R) losses in the system and improves overall electrical efficiency.

Also, operating at lower currents reduces conduction losses in the MOSFETs of the motor controller and can improve thermal performance. Lower current also allows for thinner wiring, which contributes to a reduction in vehicle weight. Higher system voltage also results in smaller relative voltage drops during operation, meaning the motor can maintain more consistent performance over the length of a race. A 48 V, 10 Ah battery, for example, stores the same energy (480 Wh) as a 24 V, 20 Ah battery. This gives us flexibility to optimize the battery pack for both energy and weight.

Despite these advantages, the transition to 48 V is not without its challenges. A 48 V pack has nearly twice as many cells in series as a 24 V pack, which increases the complexity of the Battery Management System (BMS). The BMS must monitor and balance all 13 cells, increasing the potential for cell imbalance or failure.

There are also safety considerations to take into account with all this. Although the 48 V is still classified as SELV (Safety Extra Low Voltage), it approaches the upper safety threshold and requires additional precautions like upgrading connections and more training with the system to prevent accidental shocks or arcing during operation or maintenance.

Sourcing the compatible components is another hurdle also. We would need to replace the existing 24 V motor and controller with models rated for 48 V. Higher-voltage MOSFETs and capacitors are sometimes less efficient than their lower-voltage counterparts, which may introduce minor trade-offs in controller performance. Finally, if not properly optimized, a 48 V pack could end up larger or heavier than necessary. Adding more cells increases not only the battery's size but also its structural and safety requirements, potentially offsetting weight savings achieved elsewhere in the system from changing other components.

2.3 Motors

The GEEC uses an electric motor as its sole propulsion source. Motor choice is critical in an energy efficient car because the motor's characteristics (efficiency, weight, speed, range, etc.) directly affect performance and the energy consumption. Various motor types exist, but for electric vehicles like the GEEC the most relevant categories are brushed DC motors and brushless DC motors. The GEEC currently uses a 24 V Maxon RE series brushed DC motor which has a permanent magnet DC motor as its drive unit. This high performance PMDC motor is rated for a 200 W output at 24 V, with a nominal speed of approximately 5680 rpm and a continuous torque up to 405 mNm (0.405 Nm)[7]. Notably it achieves a peak efficiency of around 94%, which shows the quality of its design with aspects like a coreless armature, strong magnets and low friction. The motor type has a profound impact on the GEEC's efficiency and design as it influences how much of the battery's energy converts to motion, how heavy or bulky the powertrain is, and what kind of control electronics are needed. In general, brushless motors tend to offer higher power to weight ratios and no brush friction losses which can improve efficiency and allow a lighter vehicle. They do require more complex controls unfortunately. Brushed motors are simpler to drive and can be extremely efficient in their optimal range but brushes introduce maintenance and speed limitations that can impact long-term efficiency. Below, a technical

breakdown of brushed DC motors is provided, an in-depth explanation of BLDC motors, and a quantitative comparison to guide motor selection for the GEEC.

2.3.1 Brushed DC Motors

A brushed DC motor, such as the Maxon RE PMDC motor in the GEEC produces torque by passing current through its coils or windings on the rotating armature. This energised rotor winding creates a magnetic field that interacts with the static permanent magnets mounted in the motor's housing or stator. The fundamental construction includes a stator where here permanent magnets are providing a fixed magnetic field and a rotor. To produce continuous rotation, brushed motors use a mechanical commutation system consisting of segmented copper commutator bars on the rotor shaft and stationary carbon brushes that press against the commutator. As the motor spins the brushes sequentially contact different commutator segments, which in turn switches the current into different rotor windings in sync with rotation. This ensures the rotor's magnetic field always chases after the stator's field, generating torque in one direction.[12]

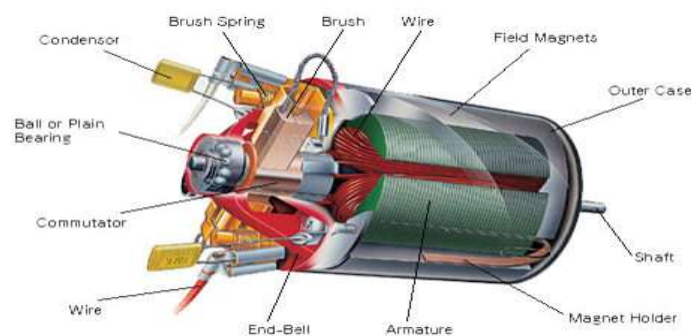


Figure 2-1 Brushed Motor Internal Structure [13]

The figure 2-1 above shows the internal components of a brushed DC motor, also known as a Permanent Magnet DC (PMDC) motor. The motor consists of a central rotating armature, around some coils of wire that are wound and connected to a commutator. An electrical current is delivered to the armature via brushes which maintain contact with the commutator as it spins. Surrounding the armature are field magnets that provide a constant magnetic field. As current flows through the armature windings, electromagnetic forces are generated, causing the rotor to turn. The shaft transfers this rotational motion to external loads, while components like the ball bearings, end bell, and outer case support mechanical stability and protection. This type of motor is widely used for its simplicity, direct voltage-speed control, and high starting torque. The rotation speed is roughly proportional to the applied voltage, and the torque is roughly proportional to the current through the motor. This behaviour can be described by the fundamental motor equations. In steady state, the motor's electrical equation is

$$V = E + I * R$$

Equation 3 Rotational Speed Of A Brushed DC Motor

Where V is the applied voltage, I is the armature current and R is the winding resistance. The term E is the back electromotive force (back-EMF) which is a voltage generated internally by the spinning armature. The back-EMF is proportional to the motor's angular velocity which is signified by the symbol ω .

$$E = k_e * \omega$$

Equation 4 Back EMF Of A Motor[14]

Where k_e is the motor's voltage constant in V/(rad/s). This is related to the motor's speed constant often given in units like rpm per volt. For example, the GEEC's Maxon motor has a speed constant of 248 rpm/V, so if 1 V is applied with no load, the motor turns at around 248 rpm and at 24 V no load, it spins at about 5950 rpm, which is consistent with the datasheet. Aswell as this the motor's torque constant k_t relates the current to the torque produced by the motor and this will be used heavily later.

$$T = k_t * I$$

Equation 5 Torque Constant of a DC Motor [15]

For the Maxon motor, the torque constant k_t is given as 38.5 mNm/A, which means the motor produces 0.0385 Nm of torque for every 1 A of current. This constant tells us how effectively the motor converts electrical current into torque. In SI units, we also use the back EMF constant k_e , which is measured in volts per (radian per second), or V*s/rad. Also in many motors including this Maxon motor the numerical values of k_t and k_e are equal when expressed in their respective SI units. So in this case both constants are approximately 0.0385, just with different units.

The equality here in these figures is not a coincidence but rather a result of how the motor is constructed. It means that the motor's ability to generate torque from current is directly related to the voltage it generates when spinning and the other way around.[16]

Using these two constants and basic motor equations, we can derive how the motor's speed depends on the applied voltage, internal resistance, and current draw known as the torque-speed characteristics. By rearranging the relevant equations, we can solve for the motor's rotational speed

$$\omega = \frac{V - IR}{K_e}$$

Equation 6 Brushed DC Motor Rotational Speed

A smaller drop in speed for each unit of torque (i.e. a flatter speed/torque slope) is generally better for efficiency because the motor maintains speed more effectively under load. Larger motors, or those with stronger magnets, typically have this flatter slope, making them more capable of handling heavier loads without slowing down. In brushed DC motors, all the windings are located on the rotor, so the way the windings are designed involves trade-offs. For example, the armature can be wound with many turns of thin wire or fewer turns of thick wire. This choice directly affects the motor's performance characteristics like torque and speed response.

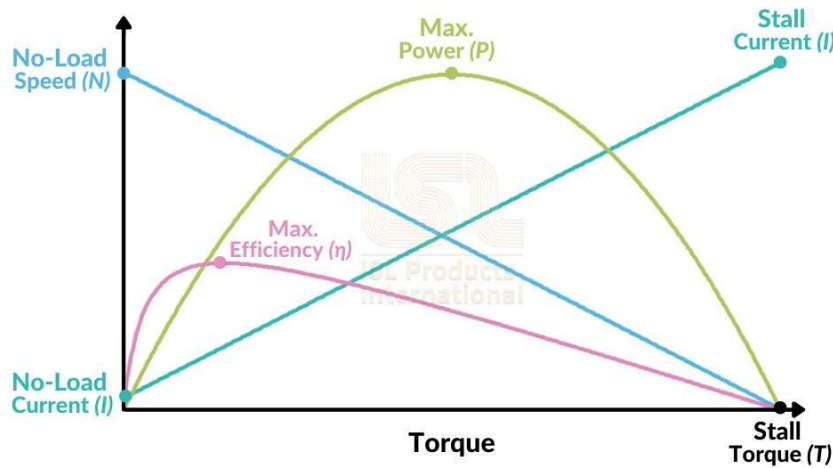


Figure 2-2 Brushed DC Motor Characteristics [17]

Using more turns of thinner wire in a motor increases the winding resistance and inductance of the motor, which results in a higher voltage constant k_e and, consequently, a higher torque constant k_t . This configuration allows the motor to operate at lower speeds for a given voltage and draw less current which effectively creates a high-voltage, low-current motor. It also means that the motor can achieve a given torque with less current, improving efficiency in some applications. However, this design tends to have a lower stall current and may reach current saturation at lower levels, potentially limiting performance under heavy loads.

Oppositely and in contrast using fewer turns of thicker wire reduces the winding resistance and inductance, resulting in a lower k_e and k_t . Motors with this configuration can handle higher currents and operate at higher speeds, making them suitable for applications that require fast rotation and higher

power output. They do require more current to generate the same torque, making them a lower-voltage and higher-current design. This setup does trade efficiency for speed and torque capabilities which is dependent on the system's power supply and thermal management capacity.

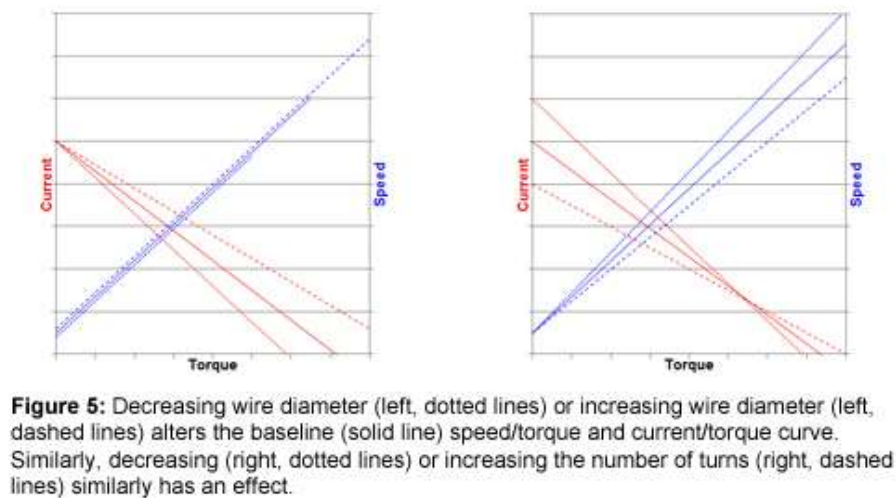


Figure 2-3 Wire Diameter & Number of Turns In a Brushed DC Motor [13]

For two motors of the same size and power rating, the one wound with finer wire and more-turns will have a lower no load speed and higher torque per amp, where the one with fewer turns can reach higher speed but draws more current for the same torque. Manufacturers often offer PMDC motors in winding variants to match different supply voltages or performance needs. The GEEC's Maxon motor, for instance, is a higher voltage winding 24 V version which is why it has a relatively high torque constant of 38.5 mNm/A and lower stall current than an equivalent lower voltage winding would have. This matching of the motor's winding to the 24 V supply helps optimise performance and efficiency.

In a brushed DC motor the commutation is handled mechanically through the use of brushes and a commutator. This makes the motor easy to operate since it doesn't require complex electronic control but it does introduce several drawbacks. The sliding contact between the brushes and the commutator causes friction and electrical losses, and is often accompanied by minor sparking. These brush losses mean that even when the motor is running with no load, it still draws a small current which is known as the no-load current. For the Maxon motor used in the GEEC this current is approximately 236 mA, which is primarily used to overcome friction and internal air resistance.

Brush friction not only causes energy loss but also contributes to heat generation and mechanical wear, which then requires regular maintenance such as brush replacement and commutator cleaning. These issues become more prominent at high speeds or under heavy load. Along with this the physical limitations of brush timing does place an upper limit on the motor's speed. At very high RPM the brushes may start to bounce or arc, disrupting commutation and reducing performance. For example, the Maxon

RE motor has a maximum rated speed of 9500 RPM, above which mechanical limitations become a concern. As a result, brushed motors are typically used in applications operating below a few thousand RPM.

Despite these limitations the efficiency of a well-designed brushed PMDC motor can be very high in its optimal operating range. Efficiency η is highest when the motor's losses (resistive, I^2R losses and friction) are a small fraction of the output power. The maximal efficiency occurs at a certain load where the current is low enough that resistive losses aren't too high but the output power is substantial.

Motor efficiency is defined as the ratio of mechanical output power to electrical input power, expressed as

$$\eta = \frac{T * \omega}{V * I}$$

Equation 7 Brushed DC Motor Efficiency [18]

This formula highlights that the motor is most efficient when it produces the required mechanical output of torque and speed while drawing the least possible electrical power. For the GEEC, efficiency is a higher priority than raw power output. The system is therefore designed so that the motor operates within its optimal efficiency range during steady cruising.

To achieve this, the drivetrain is geared to keep the motor in a high efficiency zone of the torque speed curve which is the region where $T * \omega$ is high relative to $V * I$. Instead of pushing the motor to its maximum power and maximum current, the GEEC tries and aims to stay within this efficiency sweet spot to minimize energy consumption. Speed is controlled by adjusting the voltage to the brushed motor using a DC-DC converter or a PWM controller, since motor speed is directly proportional to applied voltage for a given load. This approach ensures consistent performance while preserving battery life which is a critical factor in energy limited competition scenarios like the Shell Eco-Marathon.

2.3.2 Brushless DC Motors (BLDC)

Brushless DC motors work on the same electromagnetic principles as brushed motors by using a magnetic field interaction between a rotor and stator to produce torque but they swap the roles of the coils and commutation mechanism. In a BLDC motor the permanent magnets are on the rotor, and the windings are on the stator which is the opposite of a PMDC brushed motor. There are no brushes but instead electronic switching or commutation is used to sequentially energise the stator windings and rotate the field. Essentially, a BLDC motor is like an inside out DC motor where the electronics like the controller perform the commutation that brushes would otherwise handle.[19]

A typical BLDC motor has a multi-phase winding which is usually three phase and is distributed in the stator slots. A rotor carrying a set of permanent magnets that are usually high strength NdFeB magnets for high power density [20]. The stator windings are often arranged in pairs to create alternating magnetic poles when energised. Most small BLDCs have three winding pairs for the three phases, but they can be designed with more poles or different slot or pole counts. Removing brushes and the commutator yields a number of advantages like reduced friction from no sliding contacts, no brush dust or wear, and the ability to drive the motor at much higher speeds and power levels without mechanical sparking issues like in the brushed motors. The absence of a heavy commutator also reduces rotor inertia, and placing windings on the stator means heat is dissipated directly to the casing that also improves the cooling. All these factors allow BLDC motors to achieve very high torque and power for a given size.

For example, a 200 W brushless motor can be much smaller and lighter than the 1.1 kg Maxon brushed motor where an equivalent BLDC might weigh only around 300 g [7]. This is a four times improvement in the power to weight ratio and is a key reason brushless motors are attractive for the GEEC where every gram saved improves efficiency and on the overall score.

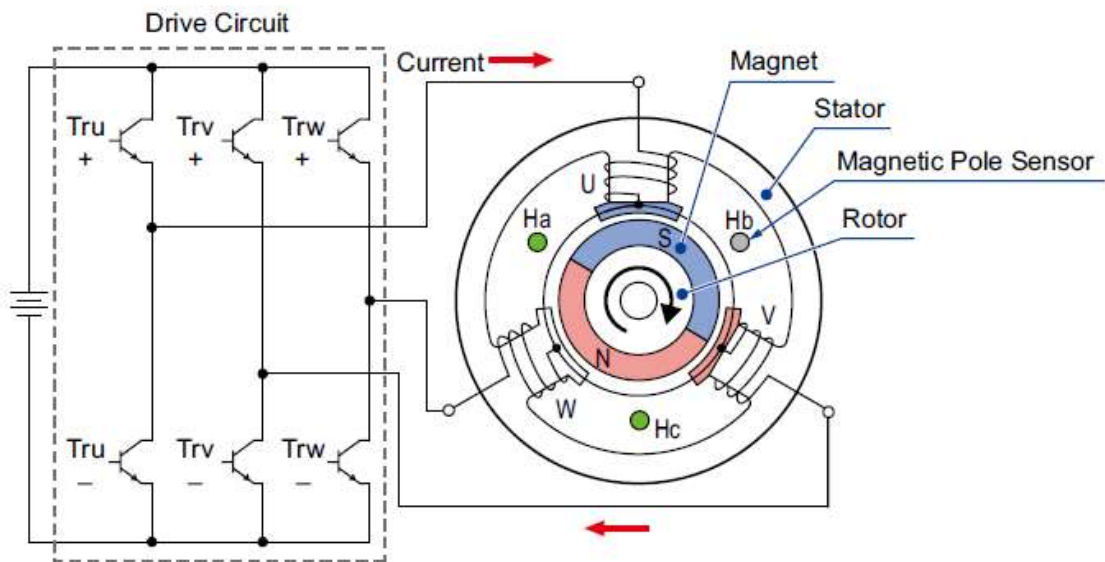


Figure 2-4 Simplified Model of a Brushless Motor [20]

The figure 2-4 above shows the internal structure and drive circuitry of a brushless DC motor (BLDC). The stator consists of three-phase windings (U, V, W), which are energised in a specific sequence to create a rotating magnetic field. The rotor is located at the centre and contains permanent magnets arranged with alternating poles (N and S). Magnetic pole sensors (Ha, Hb, Hc) detect the rotor's position

and send feedback to the drive circuit, which uses transistor switches (T_{ru} , T_{rv} , T_{rw}) to control the current through each phase. This setup enables the electronic commutation which replaces the mechanical brushes and commutators used in traditional DC motors. The result is a high-efficiency, low-maintenance motor capable of precise speed and position control.[20]

To operate a brushless DC (BLDC) motor, an Electronic Speed Controller (ESC) is required to generate a precisely timed sequence of currents in the motor windings inside the motor. The ESC functions as a three-phase inverter, which is typically constructed using semiconductor switches such as MOSFETs arranged into three half-bridges with one for each motor phase. A common design uses six transistors with two per phase. One transistor connecting the phase to the positive supply voltage (+V) and the other connecting it to ground. By selectively activating different pairs of transistors, the ESC can apply voltage across any two motor phases, while the third remains unpowered. This forms the basis of the six-step commutation method, where each switching step corresponds to 60° of electrical rotation. As the rotor spins, the controller progresses through the six steps to maintain the stator's magnetic field slightly ahead of the rotor's position, ensuring continuous motion. In effect, the ESC replaces the role of mechanical brushes, managing current switching electronically and in sync with rotor position. [21]

To determine the correct timing for commutation, the ESC must know the rotor position which can be sensed in one of two common ways. They are Hall effect sensors or back-EMF sensing. Many BLDC motors include integrated Hall sensors mounted on the stator which detect the position of the rotor's permanent magnets and provide discrete signals, to guide the ESC's phase switching. Alternatively, in sensorless BLDC designs, the ESC relies on monitoring the back electromotive force induced in the unpowered coil during each commutation cycle. As the rotor magnet passes a stator coil, it generates a voltage and by detecting when this voltage crosses zero, the ESC can infer the rotor's position and trigger the next commutation step. Sensorless control removes the need for physical position sensors and works well once the motor is spinning, though it can be less effective at startup or at very low speeds due to the absence of detectable back-EMF.

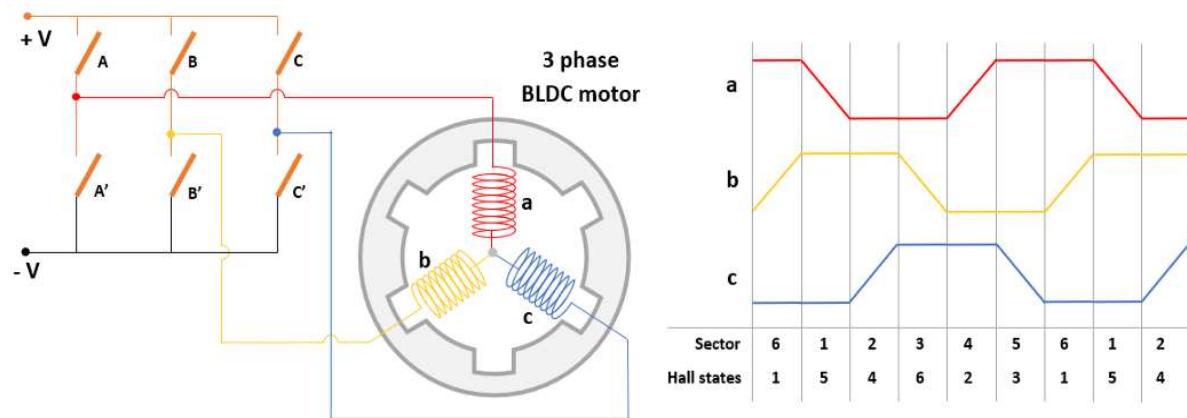


Figure 2-5 Three Phase BLDC Motor 6 Step Commutation [22]

The figure above shows the operation of a 3-phase brushless DC (BLDC) motor using trapezoidal commutation. In this method, at any moment, one motor phase is energized with a positive voltage, another is connected to ground (negative voltage), and the third phase is left floating (unpowered). This controlled switching creates a rotating magnetic field in the stator that drives the rotor forward.[22]

The left side of the figure shows the internal winding arrangement of a 3-phase BLDC motor and the corresponding switching logic used to control current flow through the phases. While the inverter switches are all shown on open in the schematic to make it simple to understand, the diagram intends to represent a particular commutation step that can be seen in the right of the figure 2-5. On the right side of the figure, the corresponding trapezoidal voltage waveforms for each of the three phases (a, b, c) are plotted over one complete electrical revolution. The waveforms are divided into six distinct segments or sectors, each representing a different commutation step. In each step, two of the motor phases conduct current while the third is inactive, forming the basis of six-step commutation. Each phase is active which is either high or low for two of the six steps and idle for the remaining four. These commutation steps are timed based on rotor position, typically using Hall sensor input or back-EMF detection as described earlier. This method of switching is known as trapezoidal commutation and is widely used due to its simplicity and efficiency. The combination of winding design, switching sequence, and rotor feedback enables the motor to deliver smooth and controlled torque which would be ideal for high-speed and low loss applications like in the GEEC.

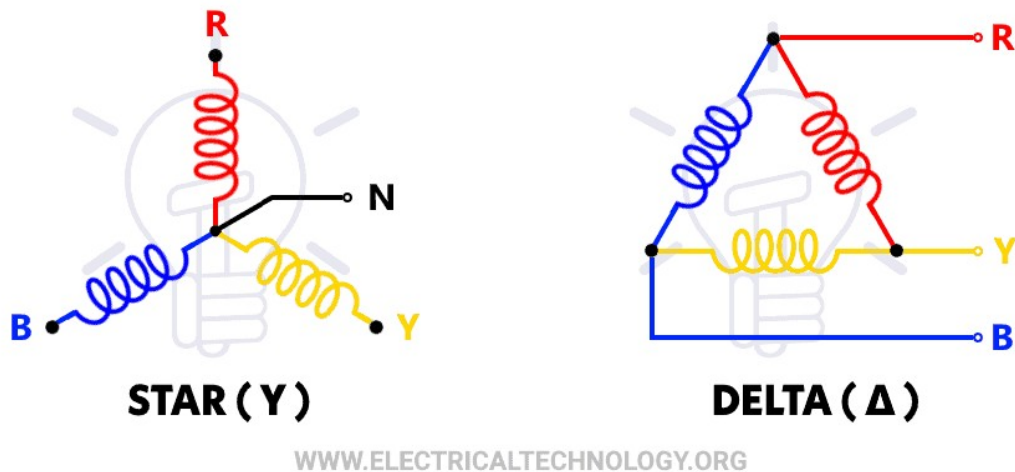


Figure 2-6 Star vs Delta Connections in Brushless DC Motor [23]

The figure 2-6 above compares the two main winding configurations used in three-phase BLDC motors which are Star (Y) and Delta (Δ). In the Star connection shown on the left, one end of each phase winding (R, Y, B) is connected to a common neutral point (N), forming a Y-shaped topology. The remaining ends connect to the ESC. On the other side where the Delta configuration on the right connects the windings from end to end in a closed loop, forming a triangle. Each junction between two windings connects to the ESC outputs, and no neutral wire is used.[23]

These topologies influence the motor's electrical behaviour and performance characteristics. In the Star each winding experiences a line to neutral voltage, which is lower and results in a slower motor speed but higher torque per amp. This configuration also reduces insulation stress and often requires a neutral line. In the Delta configuration, each phase receives the full line to line voltage that produces higher speeds, but with lower torque per amp. The Delta layout draws more starting current and imposes higher voltage across each coil so it then requires stronger insulation.[23]

In practice the Delta winding is common in applications like RC motors where maximum speed is desired, while Star winding is favoured when higher torque is needed at lower speeds such as in energy efficient vehicles like the GEEC. For the GEEC specifically, a Star-wound motor may be advantageous due to its higher torque output at 24 V, reducing the reliance on heavy gearing while maintaining efficient operation.



Figure 2-7 Inrunner vs Outrunner BLDC [24]

BLDC motors come in two primary rotor configurations that are inrunner and outrunner types. In an inrunner motor, the rotor is located inside the stator. The rotor typically resembles a cylindrical core with embedded permanent magnets and spins within a fixed outer ring of copper windings like can be seen in the figure 2-7 above. This design is structurally similar to traditional DC motors, where the outer case remains stationary while the central shaft rotates. It's compact, thermally efficient and suited to high-speed applications. The stator, being stationary and external, allows for better heat dissipation during prolonged operation.[25]

In comparison, an outrunner BLDC motor flips this arrangement to basically the opposite. The stator remains fixed in the centre while the outer casing is fitted with magnets and acts as the rotor and spins around the stator. This results in a larger rotating mass with a wider diameter that increases the torque output due to greater leverage. As a result the outrunners design excels at delivering high torque at lower RPMs. Because of their mechanical advantage, they are often used in applications like electric bikes, drones, and RC aircraft, where gear reduction can be avoided altogether.

The key trade off lies in performance characteristics between the two. Inrunners usually operate at very high speeds but produce lower torque that often requires a gearbox to be practical in applications demanding significant force such as wheel drive systems. Again in comparison the outrunners design naturally produces higher torque at lower speeds making them ideal for direct drive use cases. They may however experience slightly higher losses due to their rotating mass and air drag, and can be more difficult to mount due to the rotating outer shell.[25]

From a design perspective, inrunners are generally more compact and efficient at high RPM that makes them easier to integrate into a chassis without additional complexity. Then the outrunners design, while it's potentially bulkier and less efficient at very high speeds it simplifies the powertrain by eliminating the need for gear reduction, thanks to their high torque output. For the GEEC, using an outrunner motor might provide a practical advantage as its high torque could allow direct drive to the wheels, reducing drivetrain complexity and improving overall system simplicity.

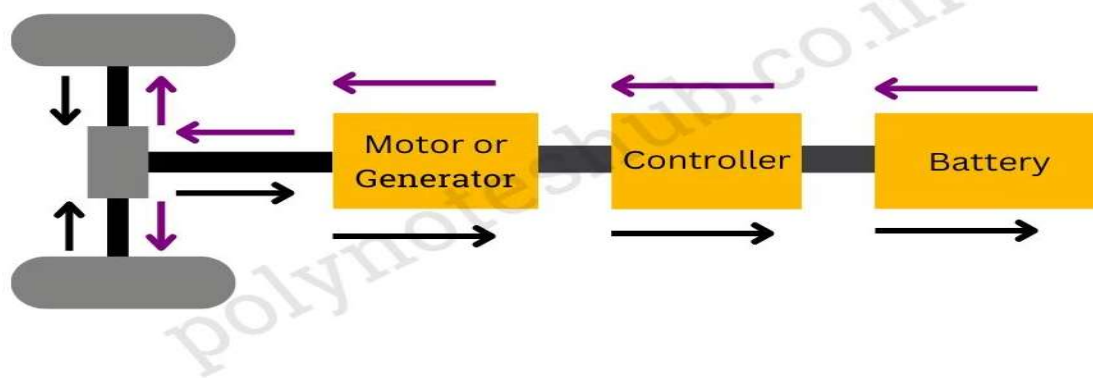


Figure 2-8 Regenerative Braking Block Diagram [26]

The figure 2-8 above illustrates the working principle of a regenerative braking system in an electric drivetrain. During normal driving energy flows from the battery to the controller, which acts as an inverter. The controller converts the battery's DC power into three phase AC signals that drive the motor. This motor which is connected to the wheels then propels the vehicle forward. In this mode, energy flows in the direction indicated by the black arrows from the battery through the controller to the motor, and finally to the wheels.

When the vehicle slows down or brakes the system then shifts into regenerative mode, as shown by the purple arrows. In this mode, the motor switches roles and functions as a generator. As the wheels continue to rotate which is due to the vehicle's inertia, they turn the motor shaft. The motor now produces electrical energy instead of consuming it. This generated energy is sent back through the controller, which rectifies the current and channels it into the battery which effectively is recharging it.[27]

Regenerative braking enables recovery of kinetic energy that would otherwise be lost as heat in traditional braking systems. This not only improves the overall energy efficiency of the vehicle but also

extends the effective range. In electric vehicles like the GEEC, integrating regenerative braking can and could be particularly beneficial during deceleration events such as cornering or any stop and go movement on the track where every joule of saved energy counts.

In summary the brushless motors offer clear advantages in power density, longevity with no brushes to wear out and often higher efficiency at high speeds. Their drawbacks are the need for a more complex controller and generally a higher upfront cost and maybe more intricate wiring. However given the increasing availability of compact, programmable ESCs linked to the motors available to be used there is plenty of info and specifications to base the building of a custom one. Due to this the implementation complexity is much less of a barrier today. BLDC motors have become common in modern EV and high efficiency drives due to these advantages. The next section provides a quantitative comparison of a brushed vs. brushless motor in metrics relevant to the GEEC.

2.3.3 Comparison of Brushed vs Brushless Motors

To determine whether the GEEC should retain its brushed PMDC motor or transition to a brushless DC (BLDC) motor, it is useful to compare key performance metrics such as speed, torque, power density, and cost at a top level first based on datasets and previous engineers work. The following sections provide a data driven comparison using figures from a broad motor survey including motors of similar scale to those used in the GEEC. These trends offer insight into the trade offs and potential advantages of adopting a BLDC system in future designs. Notably, many of the data sources and motor studies referenced here have also been cited in previous years' GEEC theses, providing continuity and validation for the design direction under consideration. [28]

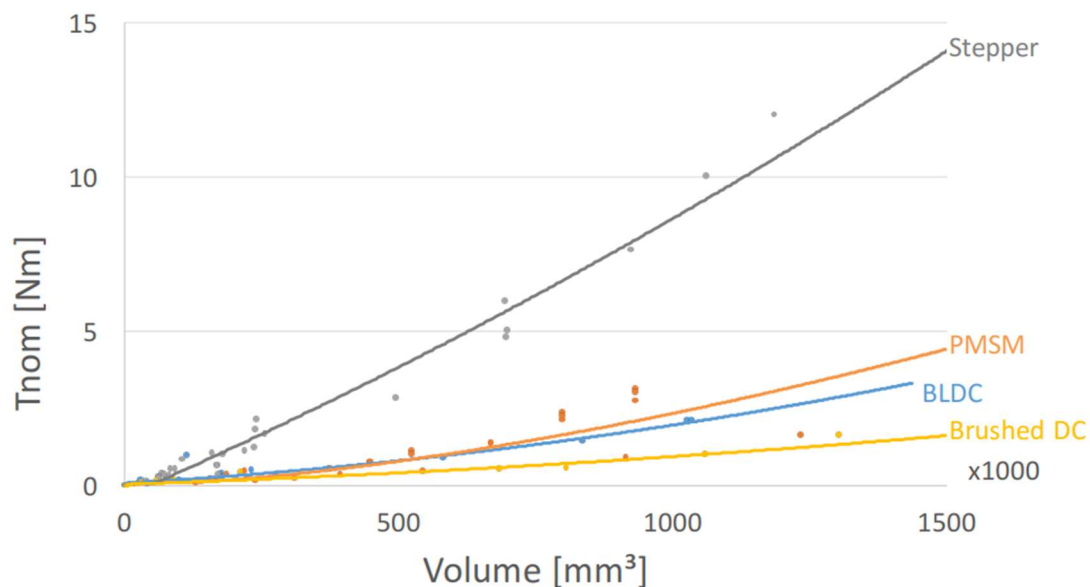


Figure 2-9 Nominal motor torque compared to motor volume [28]

Torque density is critical in compact electric vehicle designs. Figure 2-9 above compares motor torque with volume and shows that brushed DC motors offer the lowest torque per volume performance. BLDC motors outperform brushed types in this metric due to better use of space with no brushes or commutator and stronger magnets. Some BLDCs, particularly outrunner designs, can deliver high torque even with compact geometry. For instance in the GEEC's brushed motor it provides about 405 mNm in about a 0.1L volume, while a similarly sized BLDC could more than double that torque. This makes brushless options particularly appealing where space is constrained or weight distribution is critical. PMSMs and stepper motors lead in this metric overall, but between brushed and BLDC, brushless offers a clear advantage above.

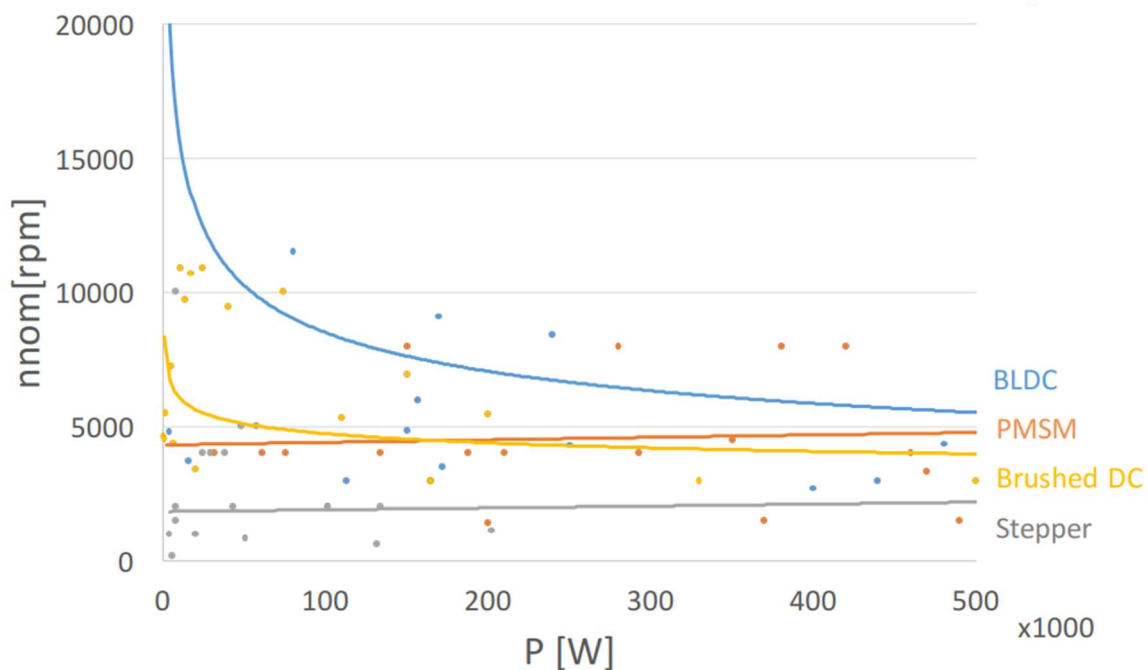


Figure 2-10 Nominal motor speed compared to rated power [28]

Brushless motors are known for their ability to achieve much higher rotational speeds than brushed motors most commonly and particularly at lower power levels. This is clearly demonstrated in Figure 2-10 above which plots the nominal speed against power. For the same rated power the BLDC motors tend to operate at higher speeds and are often exceeding 10,000 rpm while brushed DC motors typically remain below 6,000 rpm. This difference is especially relevant for applications like in the GEEC, where the current brushed motor runs at about 5,680 rpm under load. Although higher motor speeds can allow for smaller and lighter motors, they require higher gear reduction ratios to bring down the speed to usable levels for driving the wheels. That gearing adds weight and can reduce drivetrain efficiency. Therefore while the superior speed capability of BLDCs is clear, a careful balance must be struck between speed and the resulting gearing complexity. A low V BLDC or outrunner variant might be more suitable for GEEC to avoid the need for large reductions.[28]

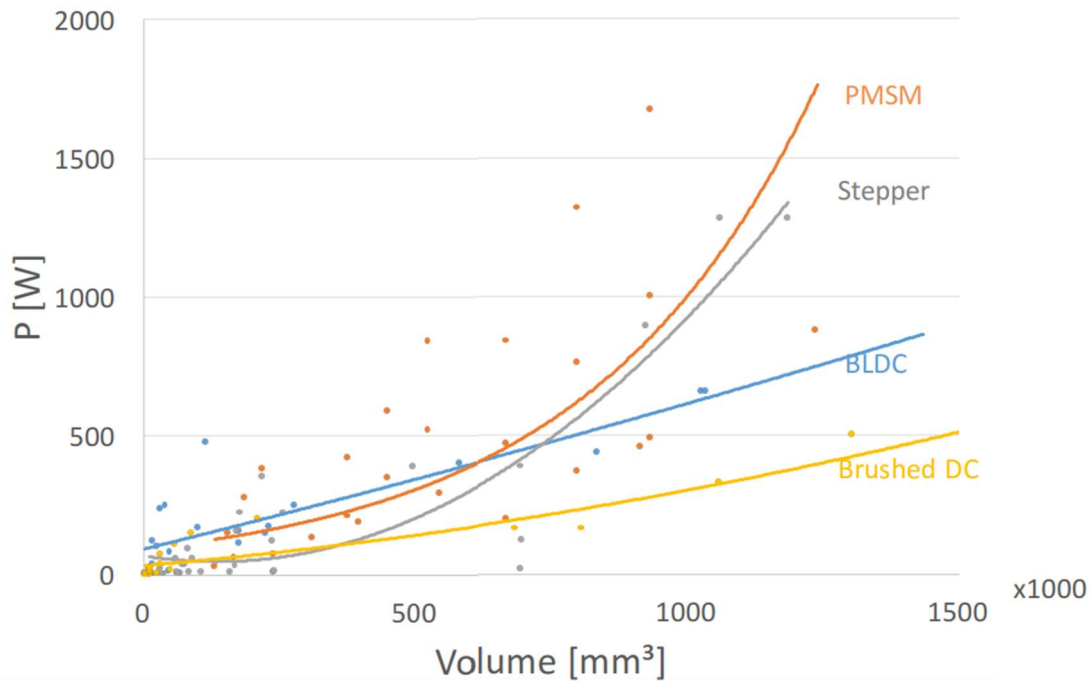


Figure 2-11 Motor power compared to motor volume [28]

Power density combines both speed and torque metrics, representing how much mechanical power can be produced per unit of size. As shown in Figure 2-11 above, BLDC motors and PMSMs have a cluster in the higher power per volume region, while brushed DC motors generally occupy the lower range. This reinforces the efficiency benefits of brushless technology already shown in sections above. The GEEC's brushed motor outputs around 200 W in about a 0.1L package, translating to 2000 W/L. In comparison, a similarly sized BLDC could deliver up to 4000 W/L, giving designers the option to either reduce motor size or increase performance without expanding packaging space. That performance margin could be especially useful during acceleration or hill climbs, while maintaining overall efficiency during races with the GEEC.

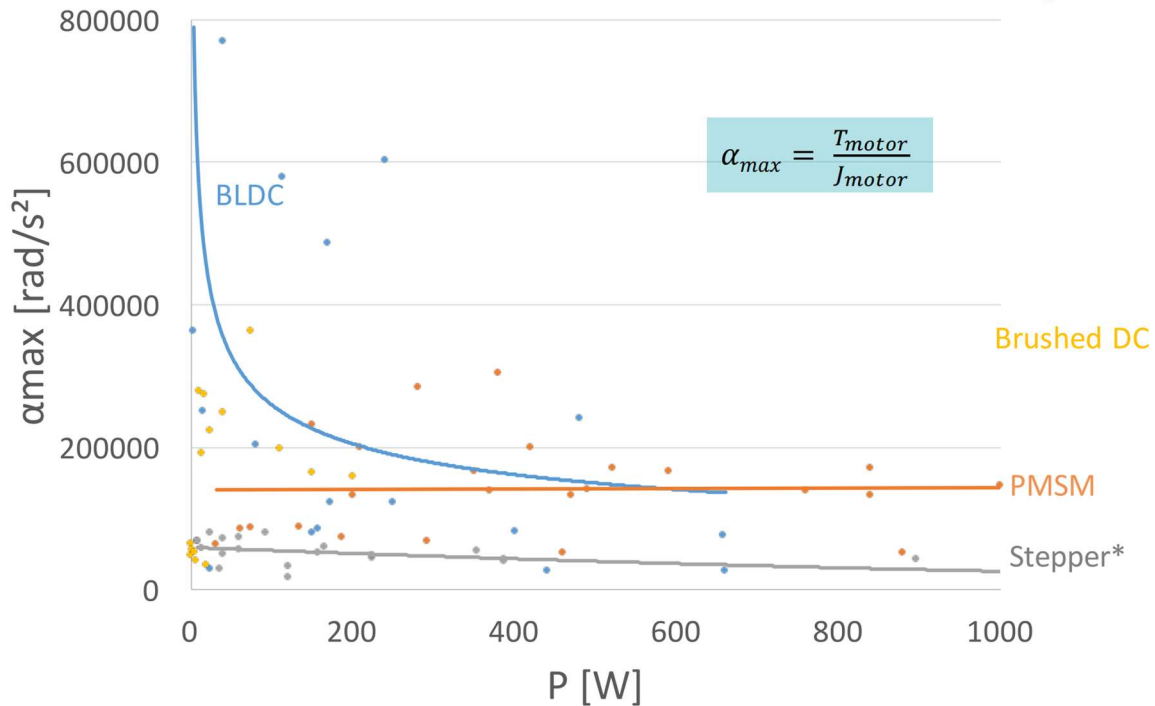


Figure 2-12 Maximum acceleration compared to motor power[28]

Dynamic motor performance is essential in applications like the GEEC, where rapid changes in speed and torque such as during tight cornering or acceleration out of turns can impact overall efficiency. Figure 2-12 highlights the superior acceleration characteristics of BLDC motors, especially at lower power levels. This advantage comes from their low rotor inertia and immediate torque response which allow for a faster adaptation to changing load conditions. Unlike the brushed motors that rely on mechanical commutation and suffer from inertia in the rotor windings, BLDC motors deliver a smoother and faster transitions. In the context of the GEEC this responsiveness translates to a more effective pulse and glide strategy and a quicker recovery from deceleration. It also has better control throughout the driving cycle making BLDC motors the more suitable choice for a dynamic high-efficiency operation.

3 Simulation & Analysis

The main goal of this project is to figure out if moving from the current 24V brushed motor setup to a 48V system, whether it be brushed or brushless, would actually give the GEEC a performance or efficiency boost. Swapping out such a critical component isn't a small decision, so the aim here is to break the system down and look at it from all angles in both theoretical and practical analysis.

This following section pulls together a mix of simulations and real world track and dyno data to help build a clearer picture. First, PSIM and LTspice are used to model the electrical behaviour at different duty cycles, focusing mainly on ripple current and what kind of impact that might have. Then, ripple data from the oscilloscope during dyno testing gives us a view of what's really happening under load. As well as just the ripple across components, analysis into switching losses is analysed, thermal properties of the current motor, and the effect of high current spikes that can occur during certain racing conditions. All these feed into a better understanding of how hard the system is working and where inefficiencies might be creeping in.

We also include a general review of the power electronics overall efficiency, check how the car performed on the track, and finally look at mechanical and iron losses which is the less obvious but things that still add up. A brief section on DC-DC converter losses is also included, since that can become more relevant if we move to a higher voltage system.

Taken all of this together, these analyses help to give a full-system perspective. By comparing this data to what we'd expect from a 48V upgrade, it's possible to decide if the change would be worth the effort and if it aligns with the overall goal of making the GEEC as efficient as it can be.

3.1 Simulation Models

To see whether switching the GEEC's powertrain and electronics from a 24V to a 48V system would offer some better performance or efficiency benefits from the current system, a series of power electronics simulations were carried out. These simulations focused primarily on ripple current behaviour across the motor at steady state operation across different motor setups and voltages. The aim was to compare the electrical characteristics of the current 24V brushed motor system against a like for like 48V upgrade under the same conditions. This kind of analysis provides early insight into whether such a change is worth implementing in future race ready GEEC designs.

Both PSIM and LTspice were used to run these simulations, but the bulk of the work was done in PSIM due to how much easier it is to customise motor parameters and modify MOSFET switching behaviour and component values. The base model for these simulations was inherited from previous years projects, but the component values were updated to match the currently used Maxon 24V brushed motor and the MOSFETs operating in the real car now. A second simulation was then run using the specifications of

a potential 48V replacement motor with comparable torque and speed characteristics. A few other motors like 36V types and different 48V candidates were also briefly considered by using their values, but they didn't show anything notably different in ripple or response, so they aren't included in the results below.

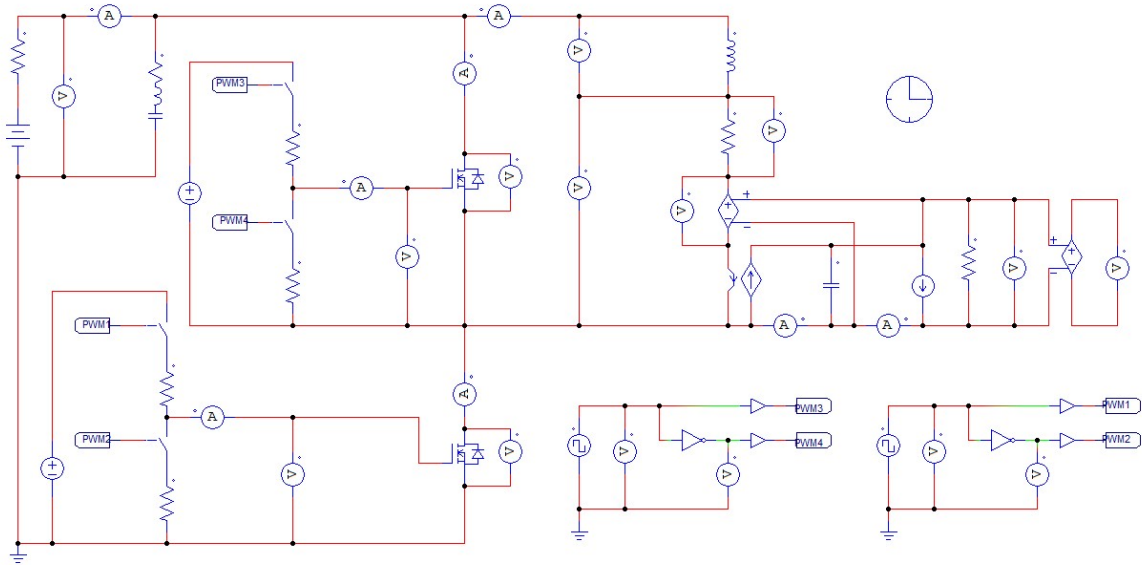


Figure 3-1 PSIM Simulation Model [29]

This PSIM model represents a simplified version of the GEEC's power electronics system, built specifically to investigate the motors behaviour under different duty cycle and voltage conditions. The structure is split into three main sections which are the power supply representing the battery, the switching stage using MOSFETs, and the motor model that simulates the mechanical and electrical characteristics of the GEEC's drivetrain.

Starting from the top left, the power supply section simulates the battery. A DC voltage source provides the system input being either 24V or 48V depending on the test and motor case. A series resistor and capacitor model the battery's internal resistance and filtering behaviour, helping to replicate realistic battery conditions. This section is crucial for observing how different supply voltages affect ripple current throughout the system.

The centre of the model contains the switching stage, which is made up of four MOSFETs like in the motor controller in the GEEC. These switches are controlled using PWM signals (PWM1 through PWM4), which dictate how current is pulsed through the motor. The duty cycle of these signals directly affects the average voltage seen by the motor, and the switching frequency influences the magnitude of current ripple.

On the right hand side is the motor model. This includes elements that replicate the brushed DC motor's internal resistance, inductance, and back-EMF generation. Additional blocks are used to simulate the speed and torque characteristics and mechanical load. Current flowing through the windings is monitored here, as this is the main output of interest when analysing ripple effects.

Overall, this model is designed to be modular and easy to modify. Key parameters like voltage, MOSFET properties, and motor specs can be adjusted to reflect either the current 24V GEEC setup or potential upgrades such as the 36V or 48V system. This makes it an ideal testbed for comparing system behaviour under different configurations.

3.1.1 24V Brushed System Simulation

The ripple current behaviour of the 24V brushed system was analysed using a PSIM model that replicates the GEEC's actual drivetrain. The simulation incorporates the maxon brushed 24V DC motor currently used in the car along with a representative MOSFET switching configuration like what's in the car. By inputting accurate electrical parameters for both the motor and switches, the model gives a realistic estimation of ripple current produced during steady state PWM and duty cycle operation. The aim of this analysis was to establish a performance baseline before considering any upgrades particularly to a 48V system.

The first result from the simulation from figure 3-2 captures the motor current over a long time period. The y-axis shows current in amps, while the x-axis represents time in seconds. The waveform demonstrates that once the initial acceleration stabilises, the system reaches a consistent operating condition. There is no evidence of instability or oscillation, which suggests that the switching configuration is behaving as expected and the simulation setup is functioning properly.

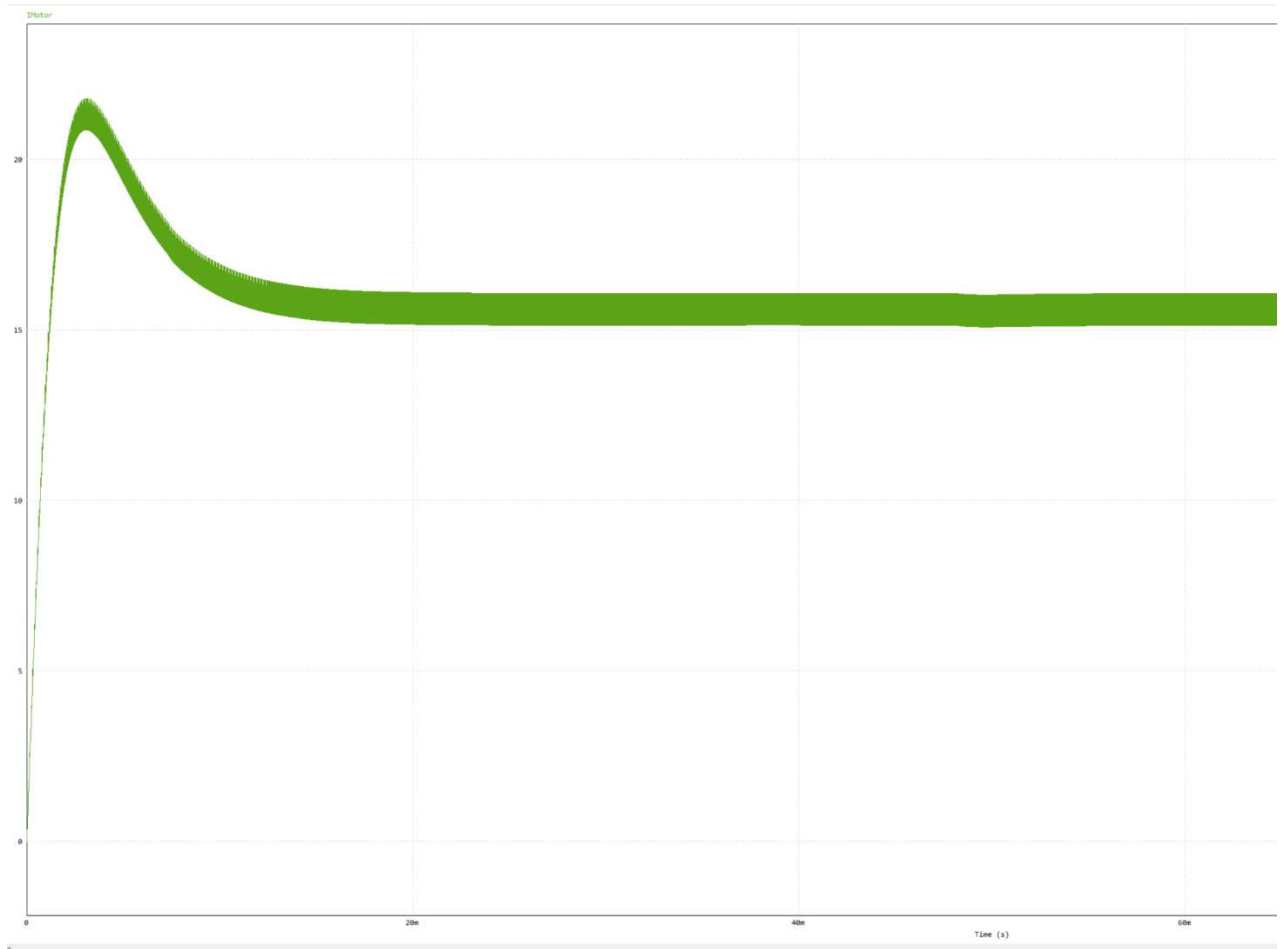


Figure 3-2 PSIM 24V Ripple Current High Level View

To better analyse the switching behaviour and assess the sharpness of the ripple, the waveform was zoomed in over a smaller time window, shown in Figure 3-3. This view captures a clean, periodic rise and fall in current, characteristic of a stable PWM and duty cycle under load. On the vertical axis the current is plotted in amps, while time is plotted along the horizontal axis in seconds. From this closer inspection, the peak to peak ripple current was measured at around 650 mA. This is a more substantial number than initially expected and could contribute to measurable I^2R losses in the motor windings. While the shape and consistency of the ripple suggest that the switching is stable and not introducing noise or instability, the size of the ripple is something that may warrant attention if a higher efficiency motor is to be selected. Apart from that, it's still within reasonable bounds for a brushed system operating at this voltage level.

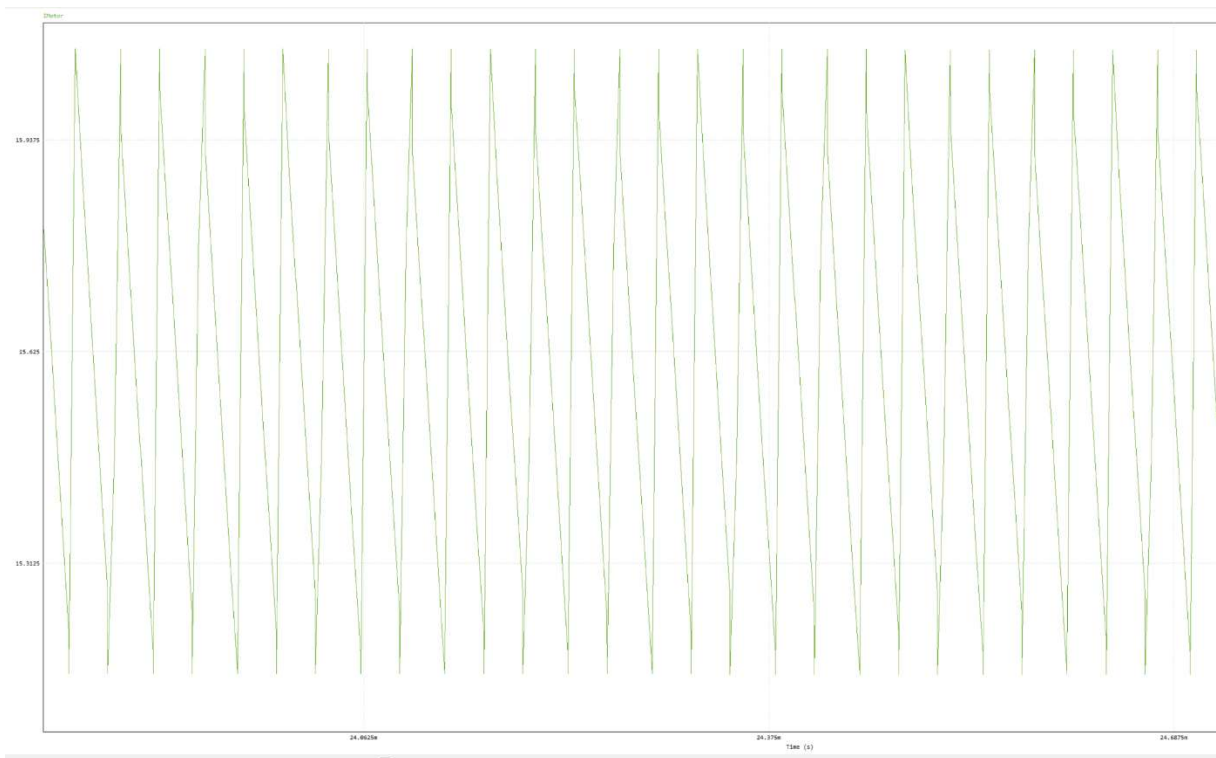


Figure 3-3 PSIM 24V Ripple Current Across Motor

3.1.2 48V Brushed System Simulation

To compare against the current 24V setup, a like for like 48V motor was simulated using the same PSIM model structure as used for the 24V case. The MOSFET characteristics and switching configuration were updated accordingly to suit a 48V system. This setup allows a clear analysis of how ripple current behaves in a higher voltage system under the same steady PWM operating conditions. Just like with the 24V case, the objective was to assess whether ripple levels remain manageable and to evaluate the efficiency implications of stepping up the voltage.

The broader timescale result from the 48V simulation again confirms a steady state PWM operation. The current response stabilises quickly and there is no sign of instability or poor switching behaviour. The steady state current output settles at around 7.8A, which is significantly lower than the 24V equivalent. This supports the general expectation that doubling voltage reduces required current for the same power output and has a direct impact on cable sizing and conduction losses.

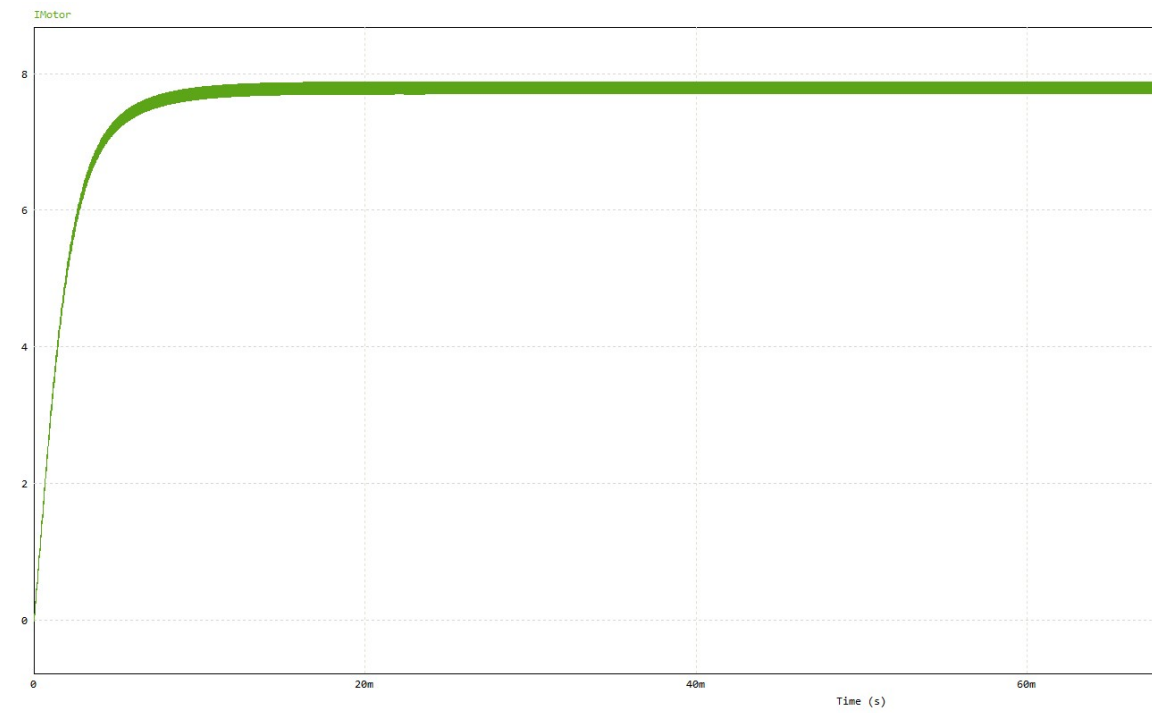


Figure 3-4 PSIM 24V Ripple Current High Level View

To better assess and analyse the switching behaviour in the 48V motor simulation, a zoomed-in view of the ripple current was taken that focused on a steady state portion of the waveform at 95% duty cycle. In this case, the current on the y axis and time on the x axis help clearly visualise the triangular ripple shape caused by PWM modulation. The waveform shows a consistent and clean profile with a peak to peak ripple current measured at around 175 mA. This is a significant reduction from the 24V simulation, which showed a ripple of around 650 mA. The lower ripple in the 48V case can be attributed to lower current draw and more efficient power delivery, and it remains well within acceptable bounds for typical

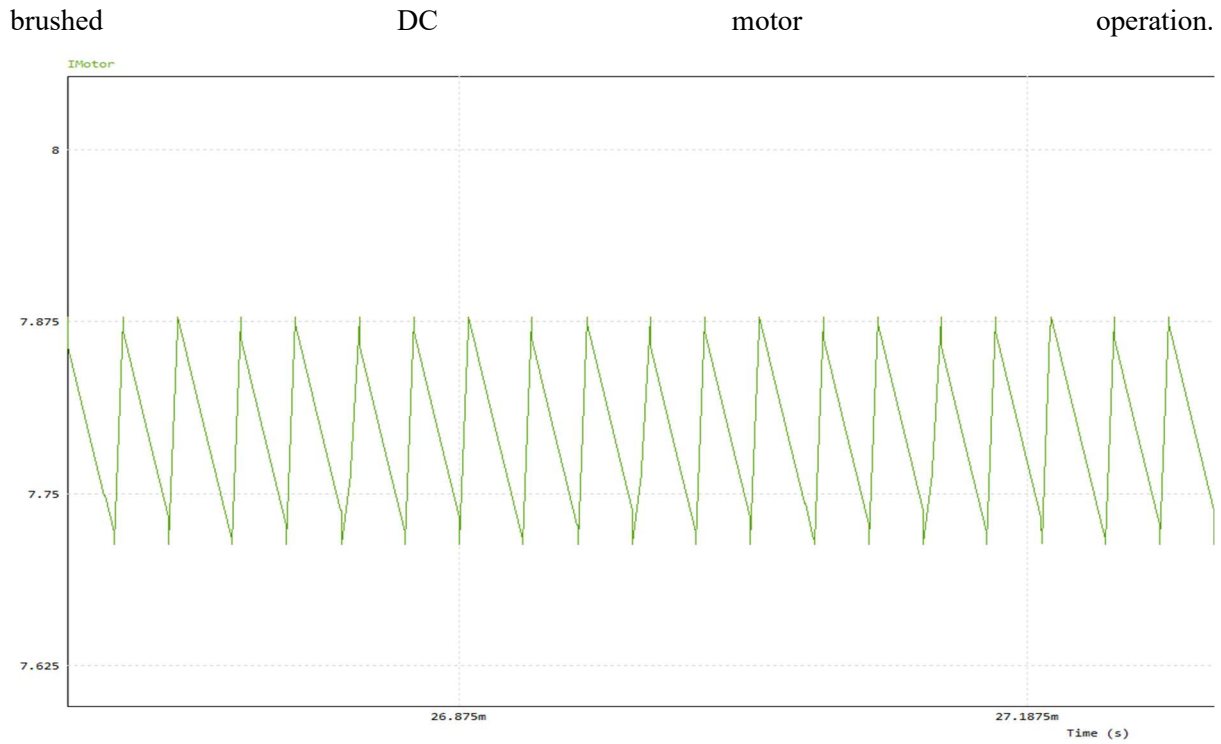


Figure 3-5 PSIM 48V Ripple Across The Motor

Despite the significantly lower ripple in the 48V simulation, the average current drawn by the motor is also lower compared to the 24V system. This results in reduced overall I^2R losses in the windings, which is beneficial for both thermal performance and system efficiency in this setup. The cleaner switching profile and lower current draw suggest that the 48V configuration could offer improved electrical performance under typical load conditions of the GEEC. With this, while the simulation shows stable behaviour, these findings would need to be verified through physical testing on the dyno and during race scenarios to ensure that no unexpected issues arise during real world operation.

3.1.3 Dyno Testing for Ripple Currents

To get real world data, tests involving the car on our dynamometer were setup. The GEEC was setup on a dynamometer under the lowest torque load possible. The dyno applied only a small resistive torque to simulate a light load, so the motor could spin near free speed. To capture electrical ripple behaviour, two oscilloscope voltage probes were attached directly to the motor terminals where one was on the positive terminal and one on the negative. This differential probe arrangement allowed measurements of the voltage across the motor to be taken, by later subtracting the + and – terminal readings. In addition, a clamp type current probe was placed around the positive supply lead feeding the motor. This current clamp measured the instantaneous current going into the motor. The goal of this setup was to record the motor's terminal voltage and input current simultaneously at high speed, using the oscilloscope so that we could observe any rapid fluctuations or ripples in these signals while the motor was running on the dyno. By using a low torque setting on the dyno meaning the motor was not heavily

loaded, it was ensured the motor controller was operating in a regime where PWM switching effects would be pronounced and easy to spot in the voltage and current waveforms. In summary, the instrumented test setup allowed the gathering of accurate data of the voltage across the motor and the current into the motor in real time, providing an insight into the ripple behaviour under dynamic load conditions.

3.1.3.1 Motor Voltage Ripple Results

The oscilloscope capture of the motor terminal voltage clearly shows a PWM switching waveform. Figure 3-6 Oscilloscope Probes Motor Voltage Ripple illustrates the motor's differential voltage as measured by the two probes placed across the motor terminals. The motor is controlled via a PWM signal, causing the terminal voltage to alternate between 0 V and a higher peak value in each PWM cycle. In this test, the duty cycle was at 95%, as the dynamometer was set to minimum torque and the motor needed only brief voltage pulses to overcome internal friction. As a result, the waveform in Figure 3-6 shows sharp, brief high-voltage spikes followed by longer low-voltage periods, which is consistent with typical PWM behaviour.

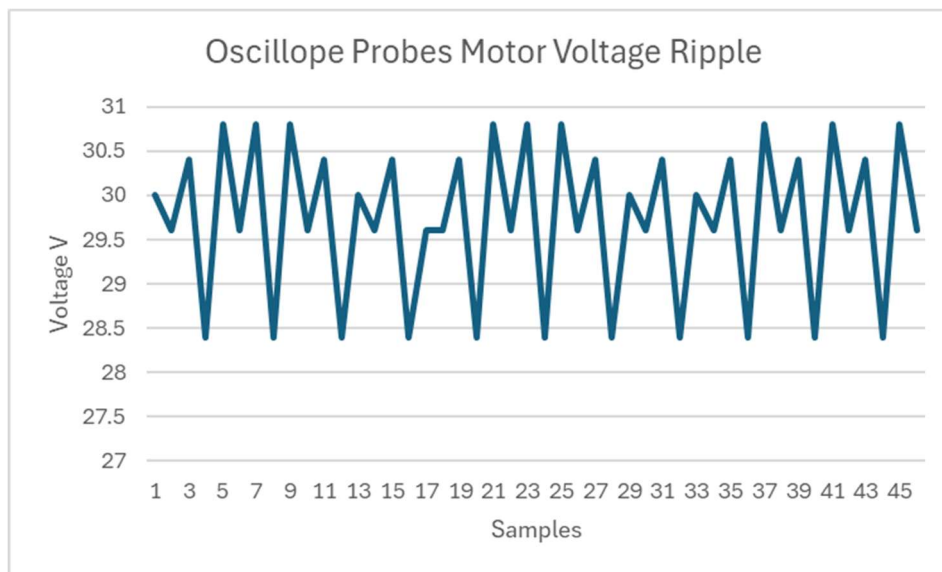


Figure 3-6 Oscilloscope Probes Motor Voltage

The oscilloscope readings report voltages peaking around 31 V, which is 3 to 4 V higher than the GEEC's onboard DAQ system recorded at the same time. This discrepancy may be due to the oscilloscope's probe calibration or small voltage offsets introduced during measurement setup. It does not invalidate the ripple pattern observed, but it highlights errors in the data. The voltage waveform still

demonstrates a classic PWM pulse train with a frequency of approximately 31.3 kHz. Minor high-frequency ringing or overshoot is also visible at the switching edges.

In contrast, the DAQ measured motor voltage shown later in Figure 3-7, captures a far smaller ripple due to its lower sampling rate and internal filtering. In the DAQ trace, the motor voltage appears relatively flat with only slight variation, as the system records an average voltage rather than capturing the full PWM waveform. This averaging behaviour aligns with expectations under low torque operations like on the dyno, where the required average voltage to sustain rotation is minimal since the back EMF nearly cancels out the applied voltage.

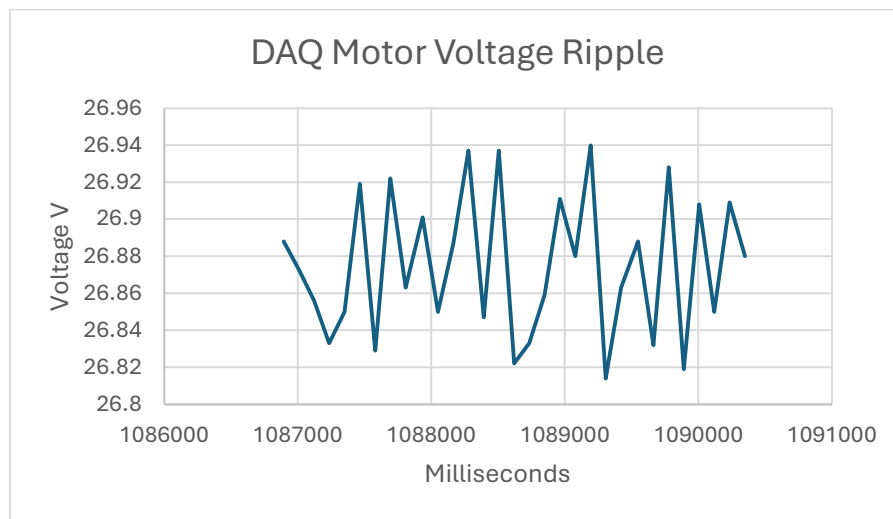


Figure 3-7 DAQ Motor Voltage Ripple

Together, Figures 3-6 and 3-7 show the contrast between high frequency oscilloscope data that reveals the true shape and amplitude of the PWM voltage ripple, and the low rate DAQ data that reflects a filtered or averaged view. This comparison emphasises the value of high speed probing when analysing electrical ripple behaviour in PWM driven motors.

3.1.3.2 Motor Current Ripple Results

The motor current is measured using a high speed current clamp reveals a distinctive ripple pattern consistent with high-frequency PWM switching. Figure 3-8 ripple analysis from current clamp shows the instantaneous current entering the motor over a short window spanning several PWM cycles. Unlike scenarios where the PWM signal is heavily modulated between high and low duty cycles, this test was conducted at approximately 95% duty, meaning the motor was continuously powered with only brief interruptions between switching pulses.

As a result, the motor current never drops to zero, but instead exhibits small and periodic triangular ripples added onto a high baseline current. These ripples arise from the rapid switching of the MOSFETs

in the motor controller, which introduce slight variations in the applied voltage across the inductive motor windings. Due to the motor's inductance, the current does not instantly follow the voltage waveform but instead ramps slightly up and down in response to each switching transition. This ramping behaviour creates the characteristic sawtooth ripple seen in the graph.

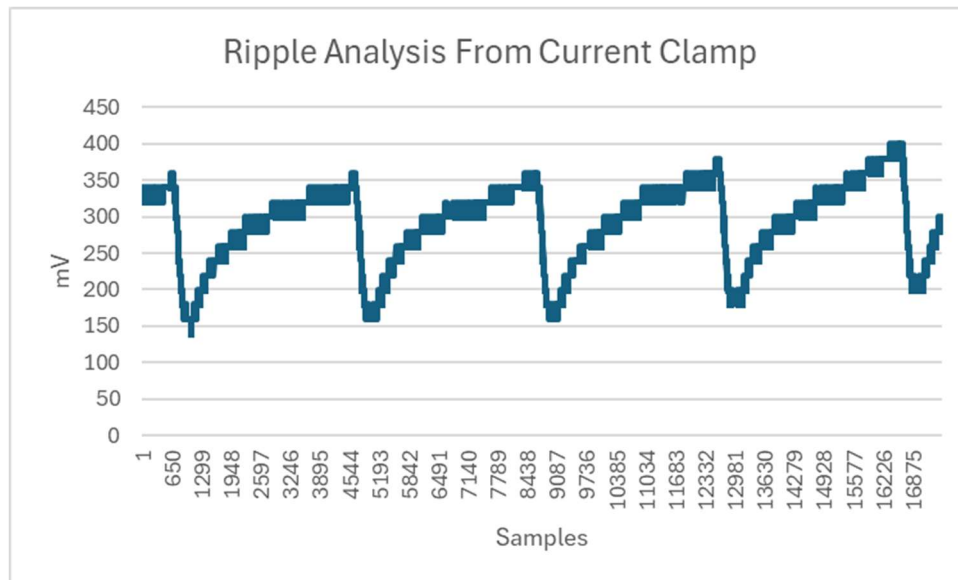


Figure 3-8 Voltage Ripple From Current Clamp

During this test, the average motor current was in the range of 3.4 A, with ripple oscillating approximately + or - 1 A around this value, resulting in a peak to peak ripple current of approximately 2 A. While this represents about 59% of the average current, it is still significant from both an efficiency and thermal perspective. The ripple increases copper losses due to the nonlinear relationship between current and resistive heating [30], and over time it can contribute to elevated temperatures in the motor windings causing even less efficiency in the motor. With such a ripple, if sustained under higher torque conditions, it may increase stress on the power electronics.

This measurement confirms what in the simulations section was previously discussed and aligns well with expectations for a system operating at high duty cycle with moderate inductance. The ripple is clearly visible but remains stable and repeatable, indicating a consistent PWM switching pattern at the 95% duty cycle, even though it still contributes to measurable efficiency losses during operation.

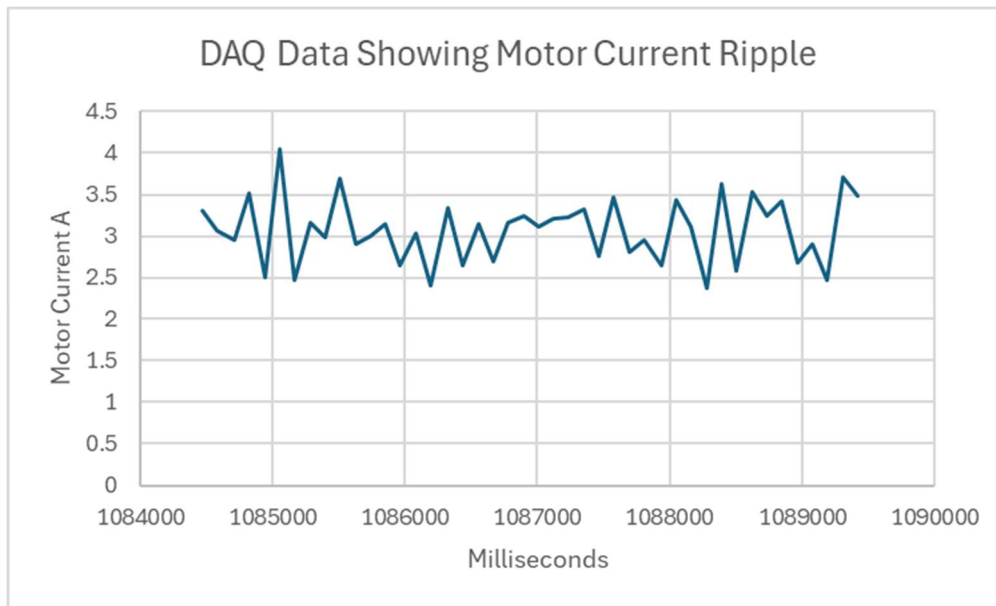


Figure 3-9 DAQ Data Showing Motor Current Ripple

Figure 3-9 displays the motor current ripple captured by the vehicle's onboard DAQ system over a short time window. While the DAQ system samples at a much lower rate compared to the oscilloscope, it still provides useful insight into the average behaviour and slower current variations in the system. The graph shows the current fluctuations between approximately 2.5 A and 3.5 A, indicating a peak-to-peak ripple of around 1 A during steady state and low load operation on the dynamometer.

This ripple is less sharp and defined than what is captured by the high speed oscilloscope, due to the DAQ's lower sampling resolution. However, the underlying switching behaviour and the effect of the PWM controller are still visible. The oscillations suggest periodic current rise and fall consistent with PWM modulation, even though high-frequency features are smoothed out.

Despite its limitations in capturing fast transients, this DAQ data helps prove the broader current trends observed in the oscilloscope measurements. The 1 A ripple shown here is basically aligned with the high speed current clamp findings, just reinforcing the presence of substantial ripple even at low torque conditions. These fluctuations are important when considering resistive losses and system efficiency over longer operating periods.

3.1.3.3 Comparison With PSIM Simulations

The PSIM simulation results for the 24 V and 48 V systems reveal a clear contrast in ripple current levels. In the 24 V model, the motor current waveform is at a steady high duty cycle at approximately 95% PWM and exhibits a peak to peak ripple of about 0.65 A, as shown in Figure 3-3. This triangular ripple is based around a high average current. In comparison, the equivalent simulation for a 48 V supply under the same conditions produced a much smaller oscillation only about 0.175A peak to peak ripple

in the motor current seen in Figure 3-5. This nearly four times reduction in simulated ripple amplitude for the higher voltage case aligns with expectations as the 48 V motor draws roughly half the current for the same power, and it likely has higher winding inductance, both factors that naturally suppress and lower these current fluctuations. The PSIM models for both voltages showed clean and stable switching waveforms, and in the 48 V case the ripple remained well within acceptable limits with a smoother current profile. These simulation findings suggested that moving to a 48 V architecture could dramatically reduce current ripple stresses on the motor and controller.

When these simulation predictions are compared to the real world measurements a noticeable discrepancy and difference arises in particular for the 24 V system. During dynamometer testing of the actual 24 V drivetrain with the motor running at 95% duty under a light load, the motor current was found to oscillate approximately + or -1A around the mean value, corresponding to about 2A peak to peak ripple as captured by the high speed current clamp seen in Figure 3-8. This measured ripple is substantially higher than the 0.65A peak to peak indicated by the PSIM simulation for 24 V, even though the test conditions were intended to mirror the steady high duty scenario. In other words, the real 24V system's current ripple was on the order of 2A which is roughly three times larger than the idealised simulation results from PSIM. Notably, the 48 V case could not be directly tested on the dyno since the vehicle currently runs on 24 V hardware and we don't have a 48V motor, but the simulations predict a much gentler ripple for 48 V. If the same real world to simulation ratio were to carry over, the actual 48 V ripple would still be expected to be far lower than the 24 V case. The dyno's slower data acquisition plot in Figure 3-9 also showed a smaller 1 A ripple for 24V due to averaging and slower sample rate, but the high speed instrumentation confirmed that rapid PWM induced fluctuations in the 24 V system are definitely more pronounced and larger than what the ideal model suggested.

Several factors explain why the physical 24 V system exhibits greater ripple current than the simulation predicted. Firstly, the simulation uses idealised components and assumptions like the PSIM motor model assumes a constant inductance, perfectly smooth commutation, and does not fully capture every outside effect. In reality, the motor's effective inductance may be lower or subject to saturation and the small series resistances and switching device characteristics can cause the current to change more abruptly. Secondly, the operating conditions in simulation vs what was in test were not perfectly identical. The PSIM model for 24 V was set up with a steady high load current where the simulated motor drew about 15 A with 0.65 A ripple at 95% duty cycle, whereas the dyno test at 95% duty cycle resulted in a much lower average current of around 3 to 4 amps because the motor was near its no load free speed. In such a light load and high duty cycle scenario, the motor's back-EMF nearly equals the supply voltage for most of each PWM cycle, meaning the current ramps up and down more sharply in short bursts. This duty cycle effect can yield a larger percentage ripple for when the PWM turns off, the current in a lightly loaded motor can momentarily decay almost to zero before the next pulse, producing

a large swing [31]. The simulation on the other hand effectively looked at a case where the motor current was continuous and high, which definitely dampened and lowered the relative ripple. Additionally the real world switching dynamics introduce differences that the simplified model may not include. For example the motor controller's MOSFETs have finite switching times and dead time intervals where the current commutation may cause brief spikes or dips. The layout inductance and any electromagnetic interference could also contribute to higher observed ripple[32]. In the PSIM environment these factors aren't totally accurate and switches operate synchronously with instant transitions, which can dampen the true ripples magnitude. Therefore, it is expected that a practical system will show more ripple than an ideal simulation, and the 2 A oscillation seen at 24V is consistent with this, even though it is higher than initially anticipated and measured.

Despite these differences in the exact values, the trend identified by the simulations is fully consistent with the experimental values. The key takeaway is that using a higher supply voltage greatly reduces the ripple current in the motor drive. Even though the 24 V simulation undervalued the ripple, it correctly demonstrated that the 48 V configuration would mitigate this and lower it significantly. The fundamental reason is straightforward where at 48 V the motor draws far less current for the same output power, and its windings which are designed for the higher voltage have greater inductance, both of which smooth the current. The cleaner current waveform observed in the 48 V simulation in Figure 3-5 points to lower high frequency stress on the system. If the GEEC's powertrain were upgraded to 48 V, one would expect the actual ripple current to drop proportionally, improving the overall electrical behaviour even under real driving conditions. Basically, the simulations serve as a valuable comparative tool which highlight the relative improvement with 48V the but real world testing is essential to quantify actual magnitudes. The 24 V dyno results underscore that practical ripple can be higher than ideal values, reinforcing and expanding on the need to incorporate safety margins and robust component choices. Crucially, they also validate and point towards the prediction that a transition to 48 V would be beneficial for ripple reduction.

3.1.3.4 Impact of Ripple on Efficiency and Motor Health

The measured current ripple during dyno testing showed a peak to peak variation of approximately 600 mA around an average motor current of 3.4 A for the lowest torque setting. This equates to the current oscillating by + or - 0.3 A each PWM cycle due to the switching of the motor controller. Although this ripple magnitude represents only about 17.6% of the average current, its implications and consequences for efficiency and motor durability are significant.

The presence of voltage and current ripple in the system has several important efficiency and thermal consequences. One of the most direct effects is increased copper loss due to ripple. Power dissipation

in the windings is proportional to the square of the instantaneous current as can be seen from the formula below.

$$P = I^2 R$$

Equation 8 Resistive Power Losses

As a result of the current fluctuations, the effective or mean square current is slightly higher than the average current alone would suggest. This means that, at the ripple peak, the motor experiences about 18% more instantaneous $I^2 R$ losses than under steady state conditions at 3.4 A. Thermally, this extra heating caused by the current ripple adds up with base copper losses and can lead to elevated winding temperatures. This efficiency loss becomes clearer when you consider how ripple affects the root mean square or RMS current. While the average motor current was 3.4 A, the presence of a + or - 0.3A ripple increases the RMS current to approximately 3.41 A. Applying the $I^2 R$ formula shows that power loss increases by around 0.7% due to ripple alone. Although this may appear minor, over long durations or under repeated load conditions, the excess heat generated from this loss builds up. Also, if ripple were larger as is often the case under higher torque or less ideal drive configurations like during the race, the loss impact scales significantly due to the squared relationship in $I^2 R$. As a result of this even the modest and small ripple levels can meaningfully degrade system efficiency over time. These temperatures can degrade insulation, reduce motor efficiency, and accelerate failure of the motor.

In terms of mechanical wear, the ripple also affects brush commutations in the motor. Brushed motors rely on continuous and consistent current flow across the commutator and brushes. Rapid changes in current, as those caused by PWM switching increase the rate of change of current or di/dt , leading to electrical stress at the contact points. These stresses can produce arcing and erosion which isn't good, especially at higher duty cycles where the ripple is narrow but still steep. Over time, this can shorten brush life and introduce reliability concerns.

The ripple induced noise and high frequency effects can also contribute to additional core losses like with eddy currents and hysteresis in the motor's magnetic materials [33]. Although these effects are secondary to copper losses, they still contribute to heating, most commonly under higher load or longer duration operation like during a race.

In conclusion, while the observed ripple current is small in amplitude, it has effects on the motor's electrical and thermal performance. These losses highlight the importance of minimising the ripple through things like improved drive strategies which will be covered in another section or a higher PWM frequency or added inductance. It could also be solved by transitioning to a higher voltage system, such as 48 V, where lower current is required for the same power, naturally reducing the ripple related issues.

3.1.3.5 Conclusion of Ripple Analysis

The comparison between the PSIM simulations and the dyno measurements makes it clear that raising the system voltage from 24 V to 48 V is an effective strategy for reducing ripple current in the GEEC's motor and drivetrain. In the simulations, the 48 V setup showed only a fraction of the ripple seen in the 24 V case, and while the actual ripple in practice may differ from the model, the underlying advantage of the higher voltage is still clear. A 48 V system just draws a lower current for the same power, which means any switching induced oscillations are smaller in amplitude. This ripple reduction has important implications as it leads to lower I^2R losses in the motor windings by improving efficiency and thermal performance and imposes less stress on the power electronics and mechanical commutation components. The measured 2 A peak to peak ripple at the 24V system represents extra heating and potential wear over time, where a comparable 48 V system and motor would substantially have less of these effects. Overall, the findings indicate that transitioning to a 48 V architecture would have a cleaner and more efficient current delivery, helping to maintain stable motor operation under and during PWM control. It also provides a greater margin to keep ripple currents within safe limits during all driving conditions during racing and testing. In conclusion, the 48 V upgrade offers a clear benefit in smoothing the motor current and reducing ripple related losses, supporting the case for adopting a higher voltage design in future iterations of the GEEC.

3.2 24V vs 48V Motor Comparison

To evaluate and analyse the real world performance of the GEEC existing 24 V brushed DC motor, a detailed analysis was conducted using data from the team's best recorded run in France in 2024. This dataset, collected via the vehicle's onboard DAQ system, captures a representative range of motor operating conditions under the competitive racing scenarios. By looking into the key contributors to power loss specifically copper losses or I^2R , frictional losses, and iron or core losses, a comprehensive understanding is gained of how electrical energy is converted into mechanical output, and where inefficiencies arise in this system. These measured losses were then used to recalculate the motor's true efficiency at different operating points, with particular focus on full load conditions during the race. This section presents the calculated breakdown of losses, evaluates their impact on performance, and outlines how motor efficiency varies with duty cycle, forming the foundation for later comparisons with 48 V systems.

3.2.1 Breakdown of Motor Losses

Copper losses, often referred to as I^2R losses, happen due to resistive heating within the motor windings. These losses are described by the equation:

$$P = I^2R$$

Where I is the instantaneous motor current and R is the winding resistance in this case, approximately 0.1 Ohms based on the maxon datasheet[7]. As the current is squared, these losses rise rapidly with increasing load making them a huge form of inefficiency under high torque or high current operation.

From the recorded data of the GEEC top performing France run, the copper losses were calculated over time, showing how they varied during acceleration, cruising, and deceleration. During peak performance at a 95% PWM duty cycle the motor current averaged between 7 and 8 A. Under these conditions, copper losses reached around 75 Watts. However, during transient events such as hard acceleration, where motor current surged up to 16 17 A, the losses increased dramatically to over 200 Watts in some cases due to high current spikes to 45 Amps.

These results align well with expectations for a brushed DC motor, where the copper losses can exceed other loss types like iron or friction under aggressive loading. Even relatively short bursts of high current significantly affect motor thermal performance and energy efficiency, reinforcing the importance of current and driving management strategies.

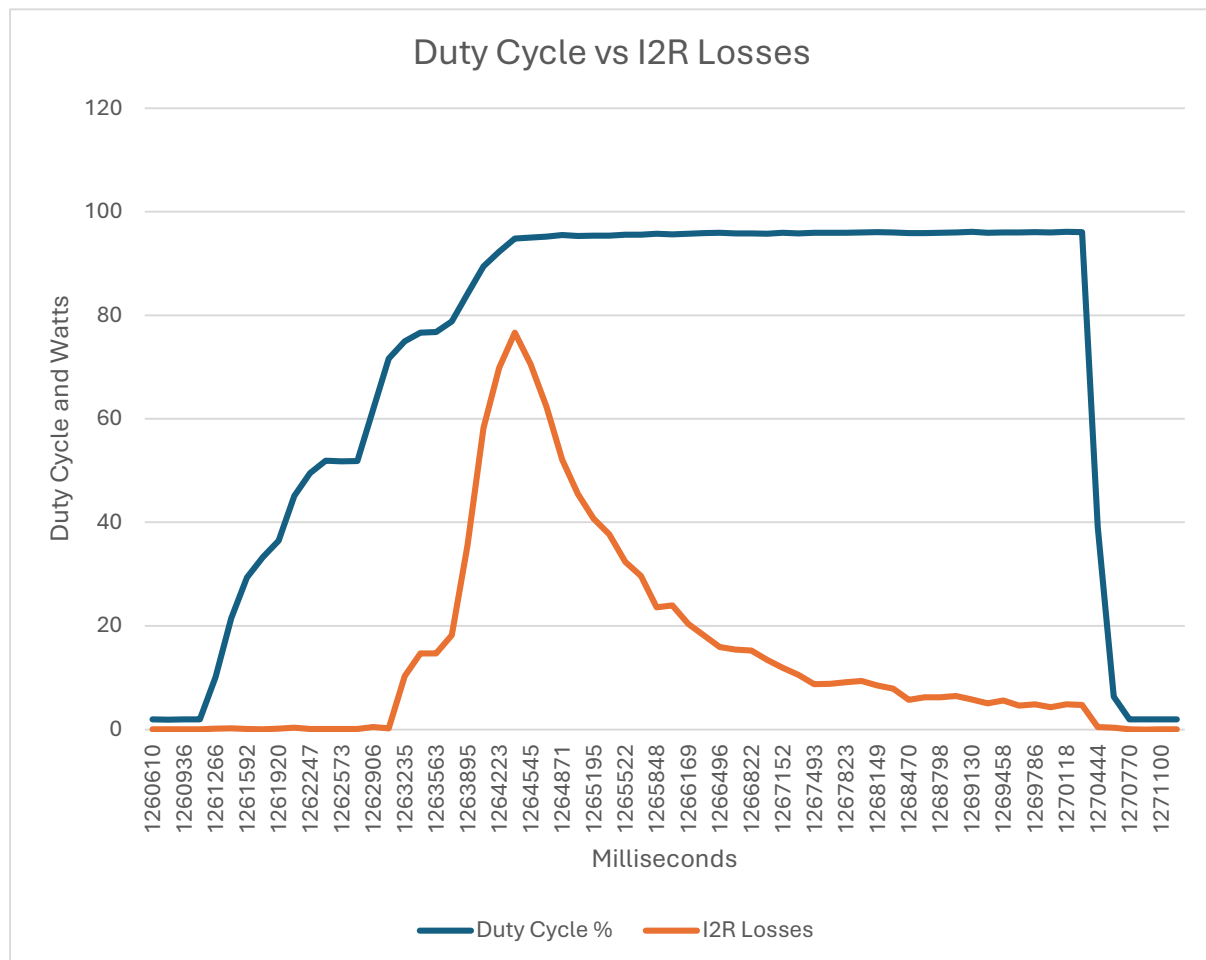


Figure 3-10 Duty Cycle Vs I²R Losses for Current Motor

Figure 3-10 above shows a time aligned plot of PWM duty cycle and I^2R copper losses. The x axis represents time in milliseconds, while the y axis shows both duty cycle percentage and copper loss in watts, sharing the same scale for ease of comparison.

In this plot, the blue trace represents the PWM duty cycle instructed by the controller, and the orange trace shows the corresponding I^2R losses. Initially, duty cycle ramps up as the vehicle accelerates, which leads to a rapid rise in motor current and associated copper losses. At peak power of 95% duty cycle, copper losses peak around 75 W. As the vehicle reaches cruising speed, the duty cycle flattens and losses taper off illustrating how resistive heating is most severe during more power hungry and intense acceleration events. This graph clearly demonstrates how higher duty cycles and higher motor current translate directly to increased copper losses.

Friction losses in the motor arise due to mechanical resistance opposing rotation, primarily from bearings and brush contact. These losses are generally small in magnitude compared to copper I^2R or core losses, but they are always present and scale with motor speed. To estimate frictional losses, the friction torque was first calculated using

$$\tau_{Friction} = I_0 * K_t$$

Equation 10 Friction Torque Calculation

where $I_0 = 0.236A$ as is the motor's no load current and $K_t=38.5mNm/A$ as is the torque constant as per the Maxon datasheet for the 24V brushed motor[7]. This gives a constant friction torque of approximately 9.086 mNm. The instantaneous angular velocity. The instantaneous angular velocity ω was then calculated using the formula.

$$\omega = \frac{2\pi * RPM}{60}$$

Equation 11 Angular Velocity Calculation

To determine the RPM, the back-EMF speed constant of 248 rpm/V was used, and the motor voltage was derived from the DAQ recorded values corresponding to each duty cycle segment. Finally, frictional power loss was calculated from the formula.

$$P_{Friction} = \tau_{Friction} * \omega$$

Equation 12 Frictional Power Loss

Figure 3-11 below shows the resulting Duty Cycle vs Friction Losses graph. It demonstrates how friction losses rise with increasing duty cycle and motor speed. At low duty cycles, friction losses are minimal, but they increase linearly with speed until plateauing near full throttle. The peak friction loss observed was around 10.2 mW, occurring at 95% duty cycle, where the motor was near its maximum operating speed during the dyno test. The trend aligns well with the expected physical model, confirming the accuracy of the approach and friction torque estimation.

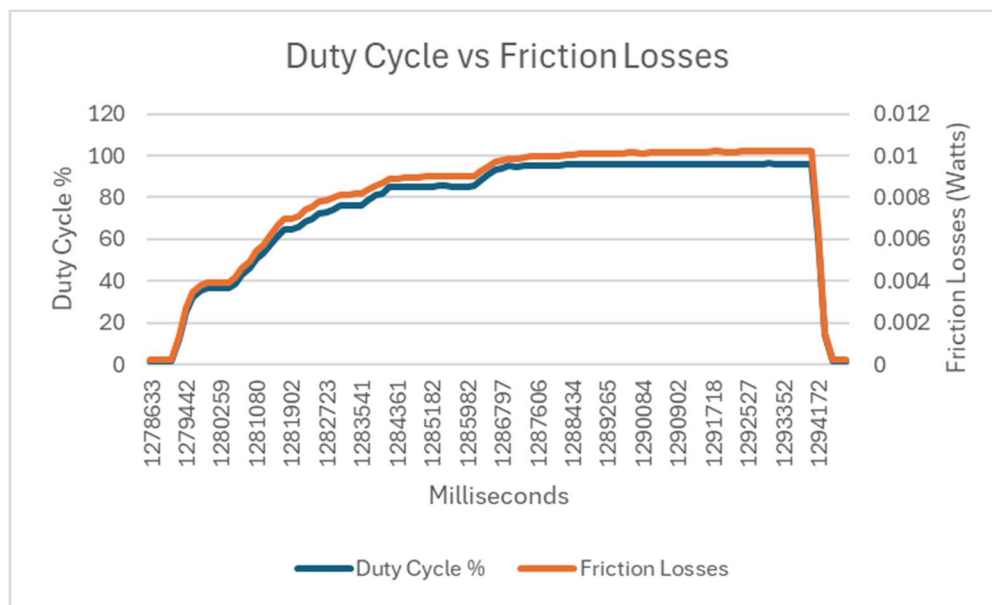


Figure 3-11 Duty Cycle vs Friction Losses for 24V Brushed Motor

While relatively small, these losses still contribute cumulatively over time to overall inefficiency of this system. More importantly, friction losses remain roughly constant with load, meaning they consume a larger proportion of power at lower operating points like at cruising or coasting. In such conditions.

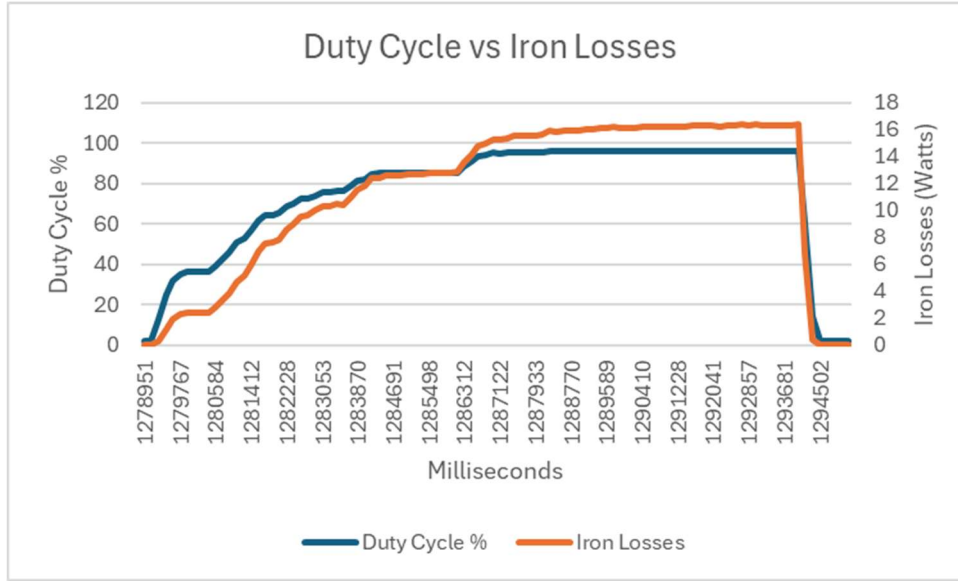


Figure 3-12 Duty Cycle vs Iron Losses in Current 24V Setup

Iron Core Losses arise due to magnetic hysteresis and eddy currents within the motor's laminated steel core as it undergoes repeated magnetization and demagnetization cycles during rotation [34]. These losses are dependent primarily on motor speed and magnetic flux variation, increasing with both frequency and intensity of the field changes.

To estimate these losses, the following formula was used.

$$P_{Iron} = K_{iron} * V^2$$

Here, K_{iron} is the iron loss coefficient derive from motor test data, and V is the instantaneous motor voltage calculated from DAQ recorded duty cycle and supply voltage. Based on nominal motor operation, a baseline K_{iron} value of approximately 0.05 was calculated which was calculated from the observed power loss at 24 V full duty.

Figure 3-12 above shows the resulting Duty Cycle vs Iron Losses graph. The trend illustrates how iron losses rise sharply with increasing duty cycle, reflecting the squared voltage dependence. At low duty cycles between 20–40%, core losses remain below 5 W and are relatively minor. However, as the motor approaches 95% duty and full rotational speed, the iron losses increase significantly, peaking around 15 W. Notably, telemetry data from full race operation has recorded core losses exceeding 60 W during prolonged high speed and acceleration segments, where voltage and speed remain consistently high.

This behaviour aligns with theoretical expectations where hysteresis loss scales approximately and linearly with speed due to frequent magnetization reversals[34], while eddy current loss grows with the

square of speed which is driven by induced circulating currents within the iron. Together, these make iron losses the dominant source of inefficiency at high speeds and a key consideration in high duty cycle operation.

The rapid rise of iron losses at high duty cycles has direct implications for the system's efficiency. Essentially, when the motor is driven near full voltage at high PWM duty cycle, a larger fraction of the electrical power is being wasted as heat in the iron core rather than producing useful mechanical work. Iron losses do not contribute to torque as they are a drain that heats the motor's laminations. Therefore, as these losses climb to tens of watts as seen in our data, the motor's output power for a given input power is reduced, dragging down the efficiency. For example, if at a certain high speed operating point the motor is drawing between 200 W to 60 W of that is going into core losses alone with additional losses in copper and friction, only the remaining power is available as mechanical output and this could easily mean the motor is a lot less efficient at that point than usual. High iron losses limits the motor's peak efficiency at top speed. This is a well known effect as iron losses tend to eat into the efficiency curve on the high speed end.

While unavoidable, such losses can be mitigated by optimizing motor materials, improving lamination design, or shifting to alternative motor types which are topics explored in the comparison and upgrade analysis sections that follow.

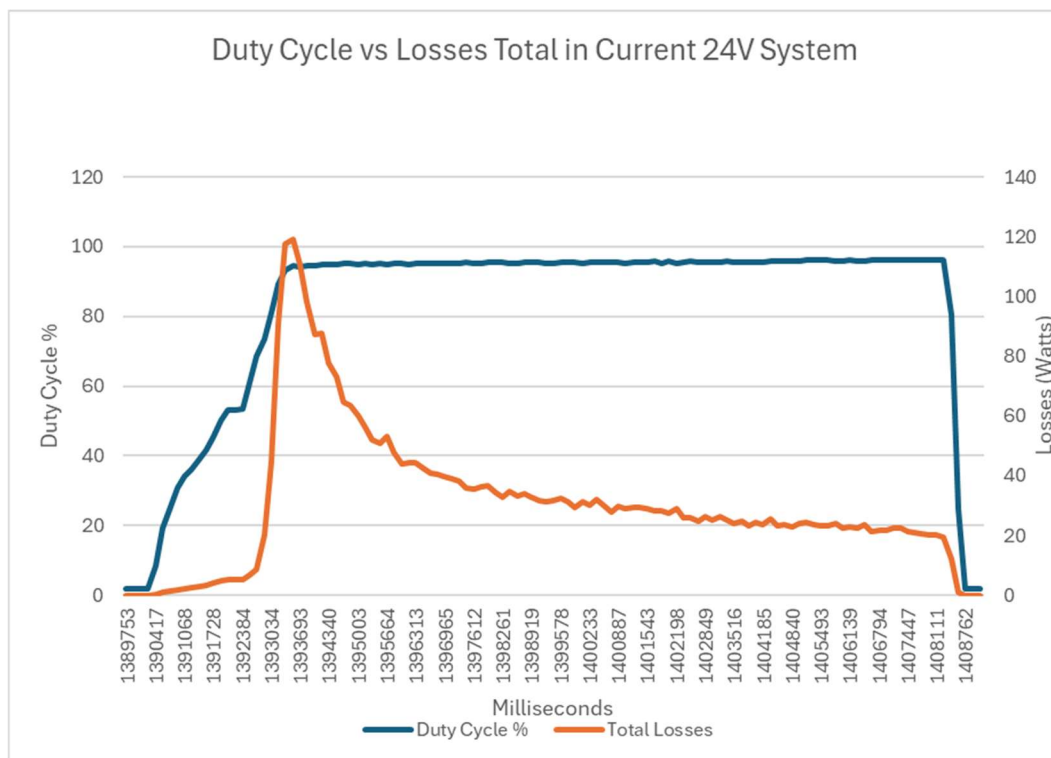


Figure 3-13 Duty Cycle vs Losses Total in Current 24V System

The total power losses in the current 24V motor system which is comprising copper, friction, and iron losses were added to assess their total impact across varying operating conditions. Figure 3-13 above illustrates the relationship between duty cycle and total losses during a representative run during the race in France in 2024. As shown, losses increase sharply with rising duty cycle, peaking just beyond 100 W at approximately 95% duty. This peak corresponds to the point of highest motor voltage and speed, where both core and copper losses are both at the same time elevated. Following this peak, losses trend downward as current demand decreases and the vehicle transitions into a cruise state. Throughout the high duty region of above 85%, losses remain consistently elevated between 20 to 30 W due to the persistent contribution of iron losses, which dominate at high speed even as torque demand stabilises. This trend highlights that while different components dominate the loss profile at different times, the combined losses still represent a significant energy drain. Accurate estimation of these total losses is essential for understanding system efficiency, battery draw, and informing targeted design improvements in future iterations of the GEEC vehicle.

3.2.2 24V Motor Efficiency

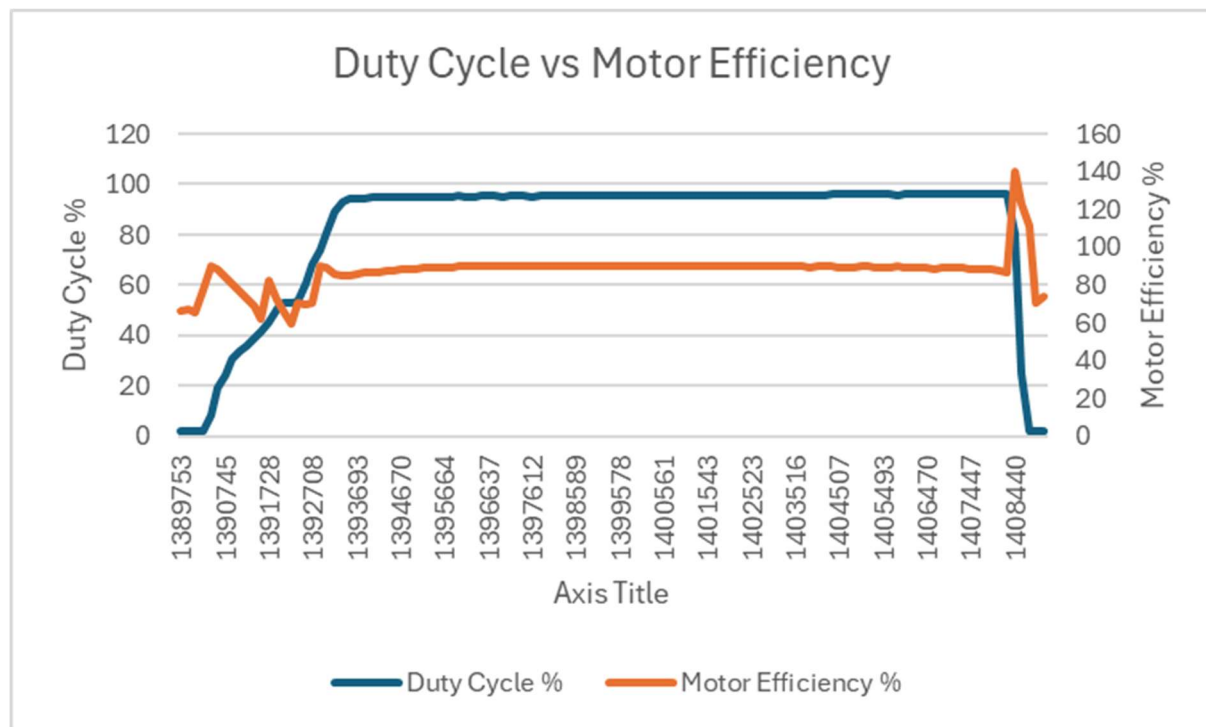


Figure 3-14 Duty Cycle vs Motor Efficiency in Current 24V System

Motor efficiency was calculated throughout the run using the following expression.

$$Motor\ Efficiency = \left(1 - \frac{P_{Losses,total}}{P_{in,motor}} \right) * 100$$

Equation 13 Motor Efficiency

where total losses include I^2R , friction, and iron losses and $P_{in} = I_m * V_m$ is the instantaneous electrical input power to the motor. Figure 3-14 shows how motor efficiency evolves relative to duty cycle across the run. While the Maxon motor used in the GEEC has a max peak efficiency of 94% under ideal conditions, the observed efficiency during this real world test averaged around 87% during the sustained high speed operating region which corresponds to 95% duty cycle. This drop is primarily due to elevated core losses at high rotational speeds and copper losses under the acceleration phases. The efficiency curve flattens in the steady state region, indicating the system has stabilised but is operating below ideal due to non negligible total losses. Deviations from the 94% ideal efficiency can also be attributed to real world factors not present in datasheet conditions such as ripple current, additional switching losses and outside actors like change in elevations. These findings highlight the importance of accounting for dynamic loss mechanisms when evaluating actual drive efficiency, especially in race conditions where sustained high speeds and power levels amplify minor inefficiencies.

3.2.3 48V Brushed Motor Comparison

Using the measured drive cycle currents and voltages and the motor's speed and torque constants, first it was determined the torque speed profile required for the France run. These values were then used to compute what battery voltage and armature current a 48 V motor would need to deliver the same mechanical torque and speed. With the 48 V motor's datasheet parameters such as terminal resistance, torque constant, speed constant and others known. The losses were recalculated with specifically the copper I^2R losses, bearing or brush friction losses, and iron core losses at each duty cycle for the same race period as the 24V motor. The resulting Figure 3-15 Duty Cycle % vs 48V Total System Losses curve plots the sum of these losses against duty cycle. Notably, at every point the 48 V system shows much lower total losses than the 24 V case. This matches the theoretical expectation to by doubling the voltage for the same output power roughly halves the current, and since I^2R losses scale with the square of current, the resistive losses fall by about a factor of four.

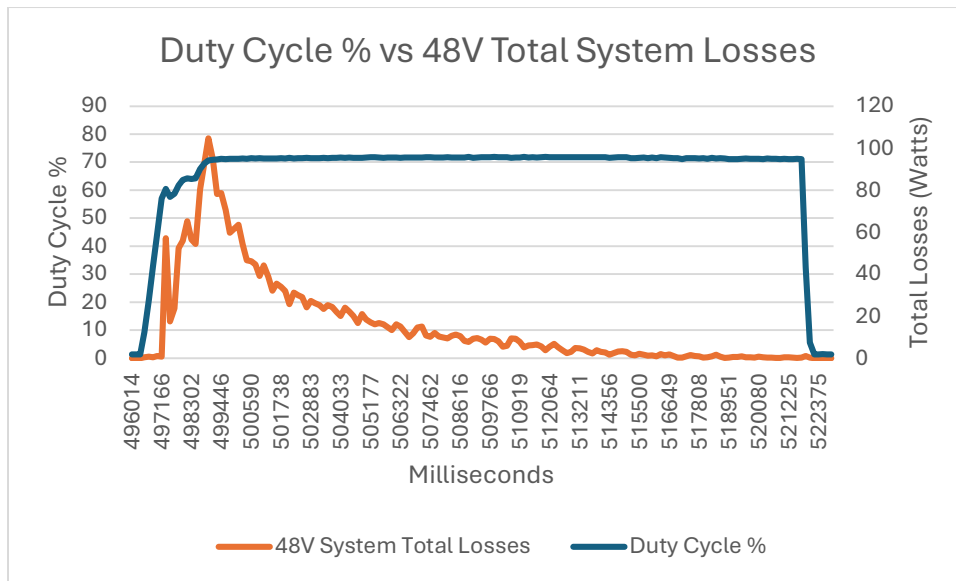


Figure 3-15 Duty Cycle vs 48V Total System Losses

In practical terms the 48V drive draws far less current for a given torque, so the copper losses and associated heating are dramatically reduced. This is consistent with general guidance that higher voltage automotive subsystems greatly reduce the current levels and allow lighter wiring with much lower I^2R losses. The loss plot above confirms that the total system losses under 48 V are substantially lower than under 24 V over the full duty-cycle range.

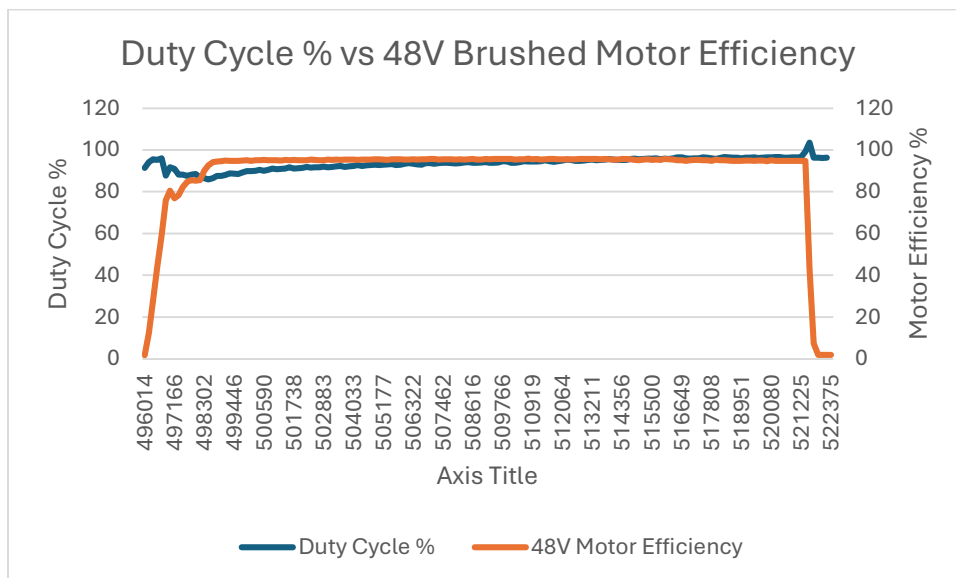


Figure 3-16 Duty Cycle vs 48V Brushed Motor Efficiency

Then the motor efficiency was calculated at each point by the same method as before in Equation 13. The Duty Cycle % vs 48V Brushed Motor Efficiency graph shows that the 48 V motor efficiency is high throughout the run with it rising to the mid 90s at high duty cycles and with an average of around

93.5% over the cycle. Comparing that to the 24 V setup that only averaged about 87% during the high duty cycle period. This efficiency gain follows directly from the loss reduction with lower I^2R losses heating under 48 V, a smaller fraction of input power is wasted. For comparison high performance permanent magnet DC motors typically run around 90% efficiency, so an average of 93% in this idealised analysis is acceptable and accurate. In other words, with 48 V the same mechanical power is delivered at lower current, so electrical efficiency rises.

Overall, the modelled 48 V scenario shows a clear improvement in motor performance. The trend is clear, the higher voltage drive produces less wasted heat and higher efficiency. In summary, the analysis above provides strong evidence that a 48 V motor setup can significantly cut losses and boost efficiency with accompanying thermal benefits compared to the original 24 V system.

3.2.4 24V Current Systems High Current Spikes

Figure 3-17 Altitude vs Motor Current presents DAQ data from the GEEC's best performance run in France in 2024, showing a clear correlation between rising track altitude and extreme current spikes. As the car ascends from approximately 89 m to just over 99 m in elevation which is indicative of an uphill section of the racecourse, the motor current surges dramatically, peaking close to 45 A. This is particularly concerning, as it approaches the threshold of the Battery Management System, risking potential tripping or system shutdown and voiding the run.

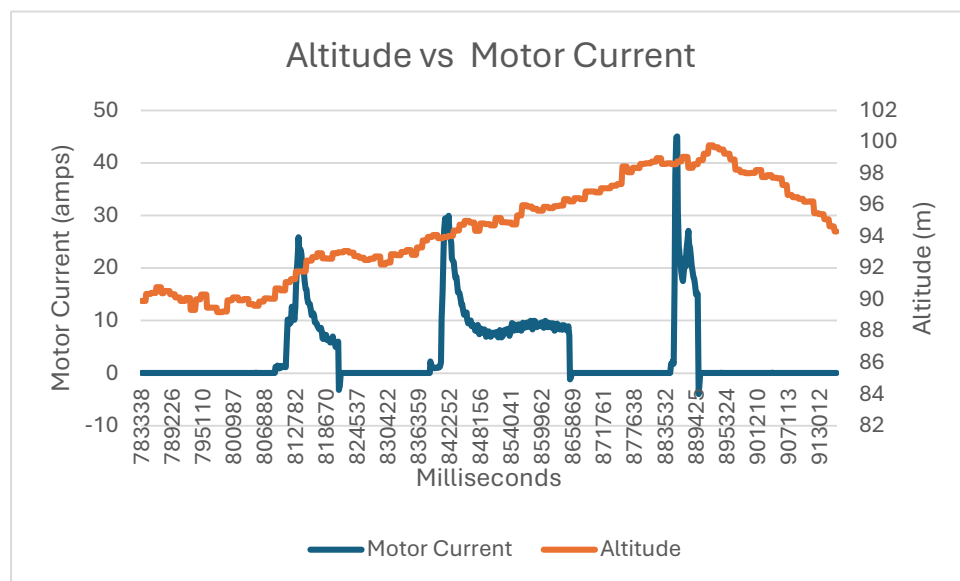


Figure 3-17 Altitude vs Motor Current in Current 24V Setup

These current spikes are due to the motor's limited torque output. When climbing inclines like this, the mechanical load on the motor increases substantially, and because the existing motor lacks sufficient

torque reserves to meet this demand efficiently, the controller compensates by driving more current into the windings which can be seen in the graph above. However, this reactive strategy only marginally improves torque while causing a disproportionate increase in resistive I^2R and thermal losses. Basically, the motor is overexerting itself in an attempt to overcome terrain induced load, revealing a critical limitation in system performance during incline acceleration.

This data strongly suggests that the current motor selection is undersized for the peak torque demands of competitive racing scenarios involving slopes. A more torque capable motor, either with a higher nominal torque rating or better torque constant k_t would allow the GEEC to climb slopes with reduced current demand. This would not only improve acceleration and maintain system stability during climbs but also significantly reduce thermal stress, copper losses, and the risk of BMS triggered shutdowns. Future upgrades should consider motors with enhanced torque profiles to better handle the full range of race track conditions.

Brushless DC (BLDC) motors, particularly those designed for high torque output, are well suited to situations where the sustained or peak torque demands exceed the capabilities of conventional brushed motors. Unlike brushed motors, BLDC motors use electronic commutation and can be optimised through controller programming to maintain high torque across a broader speed range. They also benefit from higher efficiency and better thermal performance under continuous load, which is especially advantageous when climbing inclines or accelerating under heavy demand as seen in the France run where current spikes reached 45 A. High torque BLDC motors often feature a higher number of poles, stronger permanent magnets, and better utilisation of stator windings, allowing them to deliver greater torque per ampere. Given these traits transitioning to a high torque brushless motor could significantly reduce stress on the electrical system, lower peak current draw, and improve the car's performance on slopes and during rapid acceleration.

3.2.5 48V Brushless Motor Losses and Torque Comparison

To investigate the performance advantages of a 48 V brushless motor system, the same analytical approach used for the brushed motors was extended. Using torque and speed data derived from the DAQ during the France 2024 run, the required operating conditions of torque and rotational speed at each time step were reverse engineered. These values were then used, in combination with the brushless motor's torque constant K_t and speed constant K_v , to estimate the voltage and current needed from the 48 V brushless motor to achieve identical mechanical performance.

The brushless motor selected for this study has a torque constant of 115 mNm/A, substantially higher than the 38.5 mNm/A of the original 24 V brushed Maxon motor. This represents a 3 times improvement in torque per ampere, meaning that the brushless motor can produce the same torque output at one third the current. For example, where the brushed motor may require 9 A to generate 346.5 mNm of torque,

the brushless motor would only require 3 A. This drastically reduces resistive I^2R losses, which scale with the square of current. Even though the brushless motor has a slightly higher terminal resistance ($0.236\ \Omega$ compared to $0.103\ \Omega$), the large drop in current more than compensates for this, leading to significant reductions in copper losses.

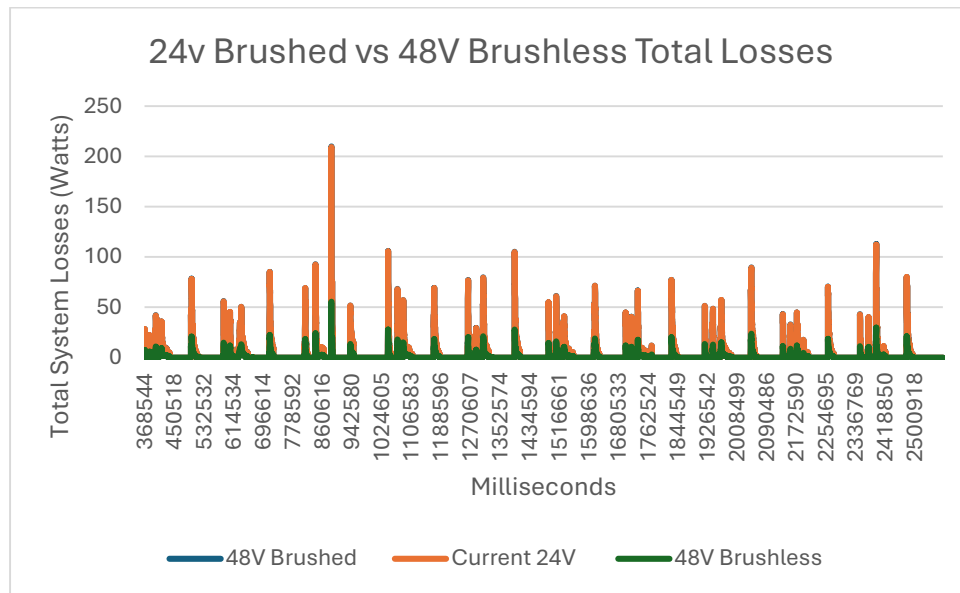


Figure 3-18 Total Losses For Different Motor Types

The effect of this improvement is illustrated in Figure 3-18 Total System Losses Comparison where 24V Brushed vs 48V Brushless vs 48V Brushed. This graph shows the total system losses throughout the same race run, calculated using identical formulas and methods for both motors. The 48V Brushed motor data is hidden behind the current 24V data as when compared to the 48V Brushless motor there isn't any notable difference between the two brushed motors. The 48 V brushless configuration results in a total energy loss of approximately 14,000 W over the entire run, compared to 53,000 W for the current 24 V brushed motor which is an improvement of over 70%. This massive reduction demonstrates how the torque characteristics and efficiency of a brushless system translate directly into superior real world energy performance.

Additionally, the brushless motor features other advantages as its lower rotor inertia of 264 gcm^2 vs 540 gcm^2 and shorter mechanical time constant 0.475 ms vs 3.78 ms vs the 24V Brushed motor contribute to quicker response and reduced mechanical losses during dynamic changes. Although the manufacturer's rated peak efficiency of the brushless motor of 91% is slightly below that of the brushed motor 94%, the real world effective efficiency is higher due to better current to torque conversion and

significantly lower heat-related losses and as shown previously the current motor is running at around 87% to 90% efficiency when active.

In summary, the combination of higher torque constant, reduced current, and dramatically lower energy losses makes the 48 V brushless motor a highly attractive upgrade path for the GEEC car. The clear drop in cumulative losses across the race run highlights the potential for improved range, thermal performance, and drive system reliability.

3.2.6 Motor Energy Consumption Comparison

The Shell Eco-marathon places a premium on energy efficiency, challenging teams to travel the farthest distance using the least amount of energy [10]. In this context, energy consumption is the most important performance metric when comparing different motor systems. Throughout testing, three motor configurations were evaluated which were a 24V brushed motor, a 48V brushed motor, and a 48V brushless motor. Different brands and motor setups were all tested along the way too and the ones with the best results made it to this report. However, due to the huge efficiency advantage demonstrated by the 48V brushless motor compared to the 24V brushed system, the focus of this section is narrowed to a direct comparison between these two configurations which also eliminated the analysis of 48v Brushed motors any further. This shift allowed for a clearer analysis of energy performance and the identification of the most suitable motor for future development.

Energy consumption data was gathered during a ten lap run on the France 2024 track. The electrical energy used by each motor was calculated using the standard formula.

$$E = \Delta t * P_{in}$$

Equation 14 Energy Consumption of an Electric Motor

where Δt represents the time step between measurements and P_{in} is the instantaneous input power. This method enabled a point by point assessment of how much energy each motor consumed over the duration of the run. The resulting comparison is shown in the chart below, where the energy consumption of the 24V brushed motor is plotted alongside the 48V brushless motor across the same time intervals for an entire run which is 10 laps.

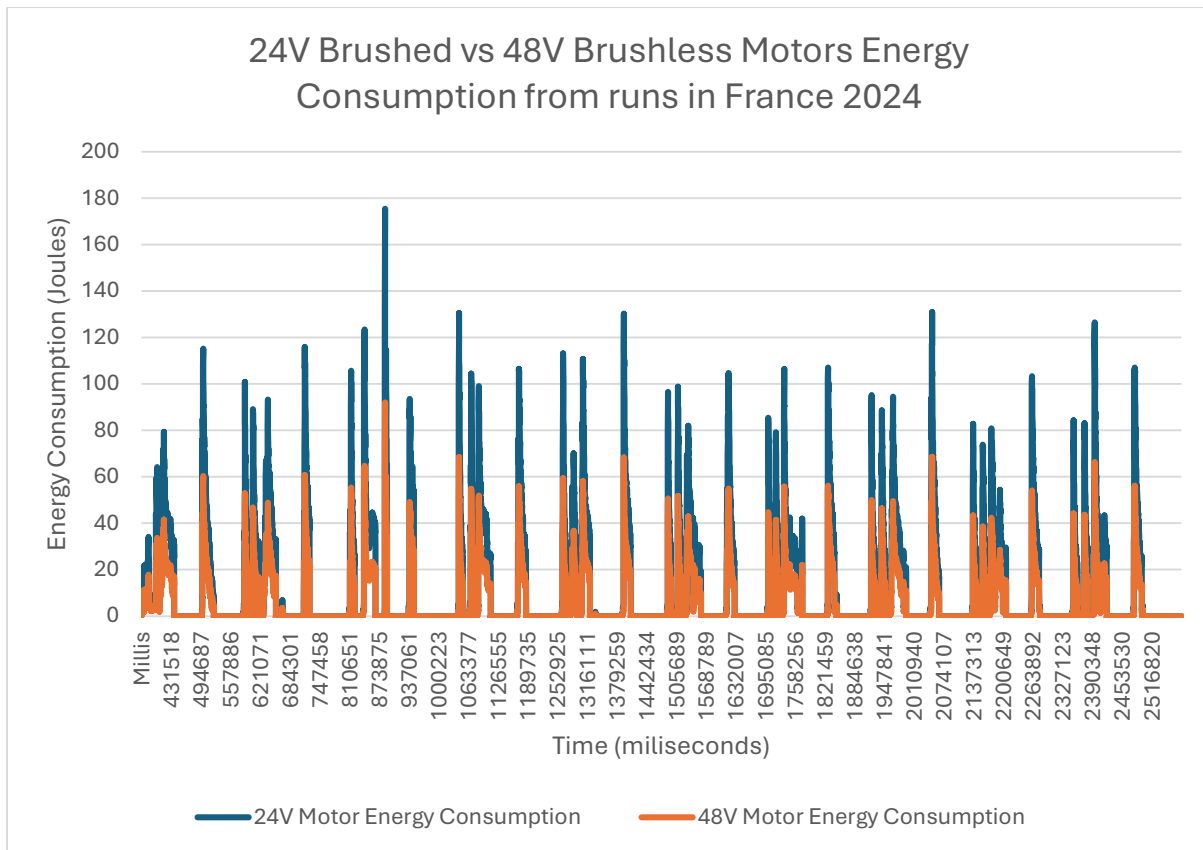


Figure 3-19 Energy Consumption Of Different Motor Types

As illustrated in Figure 3-19, the 24V brushed motor shows significantly higher peaks and more frequent energy spikes throughout the run. In contrast, the 48V brushless motor shows consistently lower energy consumption, indicating smoother and more efficient operation. The total energy consumed over the ten laps was calculated to be 163,575 Joules for the 24V brushed motor, while the 48V brushless motor used only 85,809 Joules under the same conditions. This represents a 48% reduction in total energy consumption, which is a remarkable improvement in overall system efficiency.

Several engineering factors contribute to this substantial gain in efficiency. The brushless motor design inherently avoids the friction and electrical losses associated with brushes and commutators found in traditional brushed motors [12]. Additionally, operating at a higher voltage in 48V vs 24V enables the system to deliver the same power with lower current, thereby reducing resistive I^2R losses in the wiring and motor windings. Another key advantage of brushless motors is the use of electronic commutation, which allows for more precise control over motor operation, optimising torque production and reducing unnecessary power draw across varying driving conditions.

These characteristics make the 48V brushless motor not only more energyefficient but also more stable and reliable, and more better suited to the endurance focused format of the Shell Eco-marathon. With this, this motor configuration has been selected as the basis for future development and testing. The

remainder of this report will continue to explore its integration and performance across the full vehicle system, including interactions with the drivetrain, control strategy, and energy storage.

3.2.7 Controller Comparison

As part of the transition from a 24V brushed motor system to a more efficient 48V brushless DC setup, it's essential to evaluate not only the motors themselves but also the electronic controllers that drive them. The motor controller plays a huge role in determining the overall system efficiency, thermal performance and responsiveness of the vehicle. In the context of energy optimised racing such as the in the Shell Eco-marathon, even small improvements in controller design can lead to significant reductions in energy consumption over a full run. This section compares the current brushed motor controller, with a proposed custom built BLDC controller designed to drive the Maxon EC-i 52 motor, which was the motor found to have the best characteristics based on the analysis and calculations above. This comparison focuses primarily on hardware components like MOSFETs, gate drivers, DC-DC converters, and thermal design and outlines the potential gains in energy efficiency from both improved electronics and added capabilities such as regenerative braking.

3.2.7.1 Current Controller Setup

The GEEC's motor controller uses a synchronous buck half-bridge topology to drive a brushed DC motor. At the core of this design are two Infineon IPB014N06NATMA1 60 V N-channel MOSFETs, selected to handle an upgraded 48 V system if needed. These devices offer low on resistance and moderate gate charge of about 125 nC, balancing conduction and switching losses effectively. The MOSFETs benefit from excellent thermal contact with the PCB and allow the use of heatsinks for improved heat dissipation. By implementing synchronous rectification, the controller reduces energy losses during the PWM off phase by using a low resistance MOSFET instead of a diode [35], improving efficiency over previous designs.



Figure 3-20 Infineon N-Channel MOSFET

To control the MOSFETs, the controller employs a TI LM5104 half-bridge gate driver. This IC supports high side and low side driving using a single PWM input and features built in dead time control. In addition, a LTC7004 high side driver is used to implement a soft start function for the motor. It connects the motor supply through a pre charge resistor at startup, limiting inrush current and preventing the

BMS from tripping. Once the capacitors are charged, the LTC7004 fully enhances the MOSFETs to enable normal operation. Both drivers are powered from a 12 V rail provided by a high efficiency isolated DC-DC converter.



Figure 3-21 Traco Power isolated DC-DC converter

The 12 V rail used by the gate drivers is generated by a Traco Power isolated DC to DC converter, which steps down the 24V battery input with high efficiency of around 80%. This rail supplies a low quiescent current of around 3 to 6 mA, plus pulsed current for gate charging. A separate 5 V regulator provides power to the microcontroller and sensors. Overall, the control electronics draws minimal power compared to the motor stage.

Overall the controller is compact, efficient and optimised for high frequency operation, providing a strong base for efficient brushed motor control.

3.2.8 Proposed Custom BLDC Motor Controller

For future versions of the GEEC, if the team plans to replace the brushed DC motor with a Maxon EC-i 52 brushless DC BLDC motor, which from research is the best candidate for upgrading the GEEC. Unlike the brushed setup, the BLDC motor requires a three phase inverter made up of six N-channel MOSFETs arranged in a three half bridges configuration. These devices will be selected for their ability to handle up to 60 V and optimised for high frequency switching with low gate charge and enough current capacity to sustain this system. This helps reduce switching losses while still maintaining low conduction losses. Distributing the motor's current across three phases also spreads out heat generation, allowing each MOSFET to operate cooler and more efficiently.



Figure 3-22 Maxon Brushless 48V High Torque Motor

The selected MOSFETs will prioritise fast switching over ultralow resistance, which keeps energy loss lower even at 20 to 35 kHz PWM frequencies. Their body diode performance is also important to minimise the losses during dead drive time. To drive these MOSFETs, the controller will use either three half bridge gate drivers as well or a dedicated 3 phase driver IC, which offers features like built in dead time, shoot through protection, and fault detection. This simplifies the design and ensures safe operation for this system. For example, a driver like the DGD2104 includes fixed dead time to prevent cross conduction and reduces the diode conduction loss.[36]

As with the brushed system, the BLDC controller will use a 12 V DC to DC converter to power the gate drivers. Despite doubling the number of MOSFETs, the total gate drive power remains very low at an estimated around 0.1–0.2 W. A small converter supplying 50 mA will be sufficient for this. Many driver ICs can also generate 5 V or 3.3 V rails internally for the microcontroller and sensors which would be useful. The system will support Hall sensor inputs as needed for the motor, and potentially encoders in future versions. Additional components like surge protection and filtering would be needed to ensure robust operation.

The BLDC controller is designed to be highly efficient by using active transistors like MOSFETs instead of diodes to carry current in all three motor phases. This reduces energy lost as heat. It also uses very short timing gaps at about 100 ns[4] between switching transistors to avoid unnecessary power loss through the internal diodes of the MOSFETs. The system runs at 48 V, which helps lower the current needed and reduces heat loss in the wires and components. Since the motor's current is shared across six transistors, no single part gets too hot, and the heat is spread out more evenly. Careful design of the circuit board and cooling methods like wide copper areas or heatsinks would help keep everything run cool. Overall, this setup would be expected to be as efficient or even more efficient than the previous

brushed controller, while also offering better performance, smoother motor control and the ability to use regenerative braking.

3.2.9 24V Vs 48V Batteries

In addition to differences between motor types, the system voltage itself of 24V versus 48V has a significant impact on overall drivetrain efficiency. Operating at 48V offers several clear advantages. For a given power output, a higher voltage system draws half the current compared to a 24V setup. This reduction in current translates directly into lower I^2R resistive losses in both the wiring harness and internal components, such as the motor windings and controller circuitry. Lower current also reduces voltage drops across the connectors and improves the effectiveness of the power electronics, especially when switching high currents at high frequencies. Also with this, higher voltage operation can reduce thermal strain on components, allowing the system to run cooler and more reliably. From a design perspective the 48V systems are also increasingly standard in modern electric mobility platforms, offering better compatibility with high efficiency motors and more advanced controllers. As demonstrated in this project, moving to a 48V architecture not only enabled a shift to a more efficient brushless motor, but also contributed to measurable reductions in energy consumption, smoother power delivery and improved system headroom for future upgrades such as regenerative braking and faster control loops.

4 Driving Strategy

After evaluating potential hardware improvements like the motor and motor controller, it became clear that replacing the motor and controller was not feasible within the available time. The team initially considered upgrading from the existing 24V brushed DC motor to a more efficient 48 V brushless system in order to reduce energy draw. However, integration and testing of a new drivetrain in the time left this year wasn't possible. Instead, the focus shifted to driving strategy where optimising how the car is driven in order to minimise energy consumption per lap. This decision was driven by the rationale that while hardware changes were impractical for this competition year, substantial gains could still be achieved by refining the vehicle's driving style on the track.

4.1 Driving Types

Two primary driving methods were considered for energy optimisation and from the previous years driving strategy. They were constant throttle driving and pulse and glide driving. In a constant throttle strategy, the driver maintains a steady low speed around the circuit by holding the throttle to keep the car at a near constant velocity. This approach avoids repeated acceleration cycles, theoretically minimising the energy lost in speeding up the vehicle. By contrast, pulse and glide driving involves intermittently accelerating to a target speed and then coasting with the motor off until the speed decays to a lower threshold, before repeating the cycle again. The pulse and glide method can reduce overall energy use by spending a greater portion of the lap with the motor off with zero power draw and by operating the motor intermittently at higher efficiency points during the pulses. Previous eco-marathon teams have often found pulse and glide to yield better mileage than continuous low speed running, due to lower average resistive losses and the motor running closer to peak efficiency during pulses.

Given these considerations, the strategy investigation focused on pulse driving as the more promising method, while using constant throttle as a baseline for comparison. Driving at a constant speed avoids the surge currents of acceleration, but it forces the motor to run continuously at part load which is a region where the brushed motor is less efficient. Pulse driving on the other hand, allows the motor to shut off periodically and eliminates long stretches of inefficient cruising. The expectation was that a well tuned pulse and glide strategy could outperform any constant speed approach with the given hardware.

4.2 Pulse Driving Variables

To identify the most energy efficient pulse driving technique, two key variables were examined, one being the minimum speed to coast down to before re applying throttle and two being the throttle hold duration at the peak speed. The minimum speed or reapply speed determines how much of a speed drop is allowed during coasting. For example, coasting down to 4.0 m/s versus 5.0 m/s before accelerating again. A lower reapply speed means a longer coasting phase allowing the GEEC to cover more distance

with zero power, but also requires more frequent accelerations to complete a lap. The throttle hold duration at top speed refers to how long the driver continues to apply power after reaching the target top speed, effectively maintaining that speed for a short period. A hold duration of 0 s represents the pure pulse and glide approach where cutting power as soon as the target speed is reached, whereas longer hold times of 1, 2, 5, or 10 seconds in the tests simulate a partial or constant speed cruise at the peak speed. These variables create a spectrum of driving patterns ranging from aggressive pulse and glide with no hold and deep coasting to near constant speed operation with a very long hold and shallow coasting.

To evaluate these strategies, an extensive series of tests and calculations was conducted. Acceleration profiles were first recorded using the dynamometer and through controlled track runs in November and April. These tests provided data on how quickly the car accelerates under full throttle and how much electrical energy is drawn from the battery during each acceleration. At the same time, coast to down tests were performed during a previous testing session in Edgeworthstown the year before in 2024 to measure the distance the car travels when coasting from various speeds down to a lower speed. By combining the acceleration data for the powered portion of a pulse with the coasting data for the unpowered glide portion, simulations could be made of a full pulse and glide cycle for different parameter combinations. Specifically, for each choice of top speed, reapply speed, and throttle hold time, the following metrics were determined. The energy consumed during the pulse from accelerating and holding if holding happened, the distance covered during the entire pulse and glide cycle and the total time of the cycle. Given the known Shell Eco-marathon Urban Concept track length of 1319 m this year, the number of such pulses required to cover one lap and the total energy consumption per lap were then computed for each scenario. This approach allowed us to quantitatively compare how different driving strategies would perform over a full lap without having

to physically test every combination on the track.

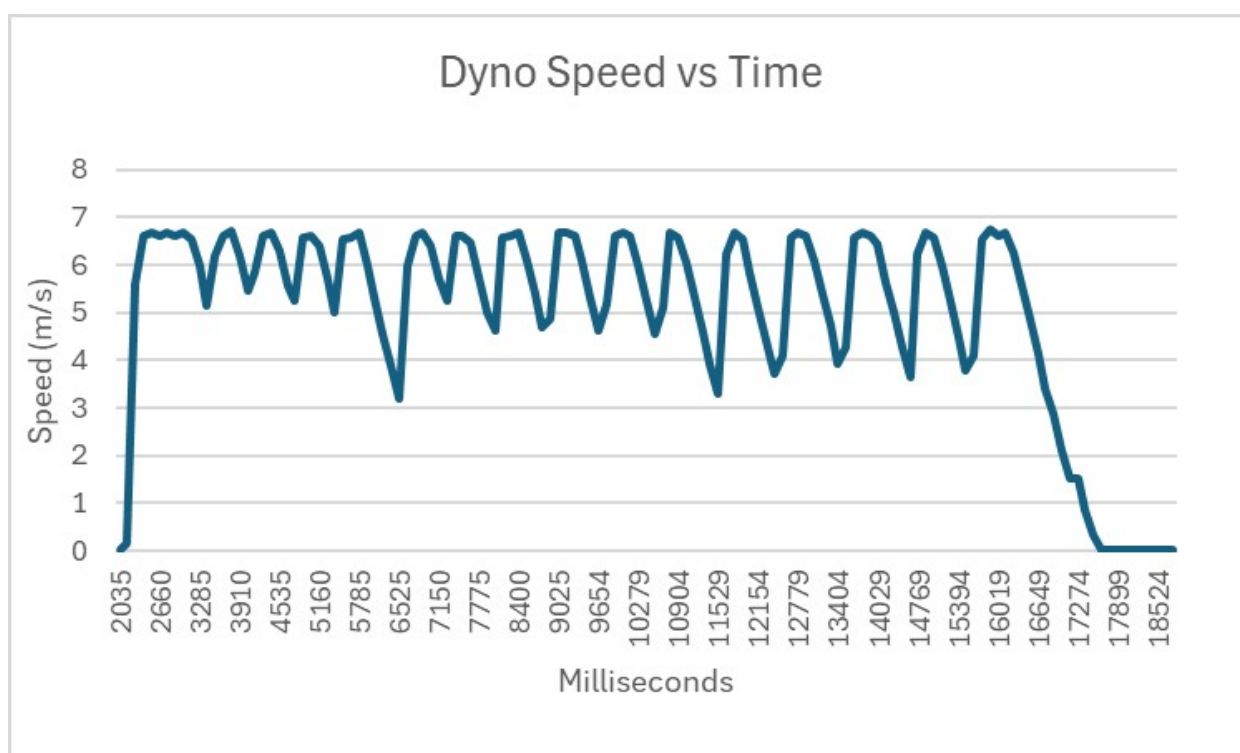


Figure 4-1 Dyno Car Speed vs Time

Figure 4-1 above shows the dynamometer speed profile illustrating the acceleration and coasting behaviour during pulse-driving tests. The vehicle rapidly accelerates from rest to about 6.5 m/s, approximately 23.4 km/h, then the throttle is cut and it coasts down, repeating this cycle multiple times. A final longer deceleration to 0 m/s is seen at the end of the test, once the series of pulses was completed. This dyno result confirms the vehicle's ability to execute consistent pulse and glide cycles, and provides data on acceleration performance for the strategy calculations.

Figure 4-2 below, shows the blue trace of the 24 V brushed DC motors distinct spikes in energy usage corresponding to each throttle on phase during the race, separated by periods of minimal energy use while coasting. Each spike's area represents the electrical energy in joules expended to accelerate the car for one pulse. For comparison, the orange trace of the 48 V brushless motor illustrates how a more efficient motor would consume significantly less energy for the same motion profile showing each orange spike is much smaller. This prior data from test runs in France 2024 highlighted the potential gains from a hardware upgrade, however with the motor swap not feasible this year, the focus shifted to extracting similar benefits through an optimised driving strategy.

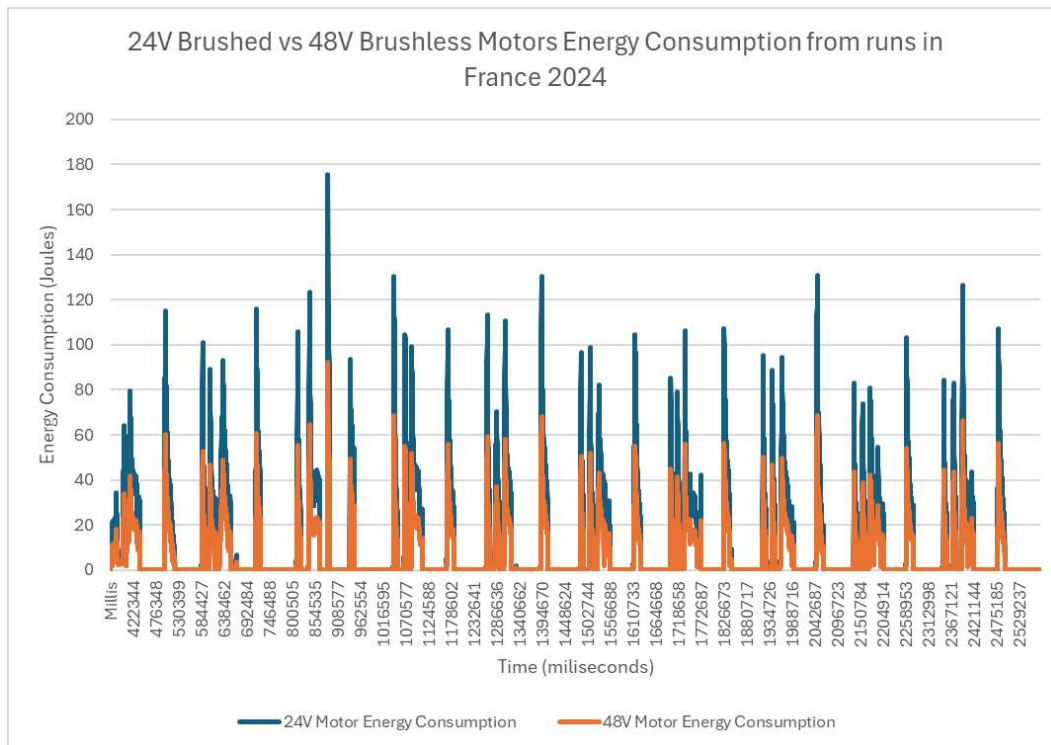


Figure 4-2 24V Brushed vs 48V Brushless Motors Energy Consumption from runs in France 2024

Table 4-1 below shows a summary of pulse driving test results for various throttle hold durations and reapply for minimum speeds. Each grouped block of rows corresponds to a different hold time of 0 s, 1 s, 2 s, 5 s, 10 s, and within each group the outcomes for three minimum speed thresholds of 5.0 m/s, 4.5 m/s, 4.0 m/s are listed. For each combination, the energy used in one pulse cycle to accelerate and hold if there is one, the distance covered by that cycle, and the duration of the cycle were determined from the dyno and coast down data. Using these values, the table also calculates the number of such pulses required to cover one 1319 m lap and the corresponding total energy per lap. The lowest energy consumption per lap achieved in our analysis is 4939J, highlighted in green, which occurs with a pure pulse and glide strategy with a 0 s hold and a 4.5 m/s reapply speed. This was then verified by completing these test profiles on the track testing completed in April 2025.

Hold duration	Decel speed (m/s)	Energy/pulse (J)	Distance/pulse (m)	Time/pulse (s)	Pulses per lap (1319 m)	Energy per lap (J)
No Throttle Hold (0 s)						
	5.0	340.0	65.92	10.8	20.01	6802
	4.5	389.2	103.94	18.0	12.69	4939
	4.0	486.4	125.59	23.5	10.50	5108
Throttle Hold 1 s						
	5.0	609.6	93.92	11.8	14.04	8560
	4.5	658.8	131.94	19.0	10.00	6586
	4.0	756.0	153.59	24.5	8.59	6492
Throttle Hold 2 s						
	5.0	879.2	121.92	12.8	10.82	9511
	4.5	928.4	159.94	20.0	8.25	7657
	4.0	1025.6	181.59	25.5	7.26	7449
Throttle Hold 5 s						
	5.0	1688.0	205.93	15.8	6.41	10812
	4.5	1737.2	243.94	23.0	5.41	9393
	4.0	1834.4	265.59	28.5	4.97	9110
Throttle Hold 10 s						
	5.0	3036.0	345.93	20.8	3.81	11576
	4.5	3182.4	383.94	28.0	3.44	10393
	4.0	3182.4	405.59	33.5	3.25	10349

Table 4-1 Driving Strategy Values

From analysing the results in Table 4-1, several clear trends emerge. First, the pure pulse and glide with no hold strategy consistently outperformed strategies that include any throttle hold. For each coast down speed tested, the 0 s hold case yielded lower energy per lap than any longer hold duration. This is because any amount of throttle hold keeps the motor powered at the top speed, incurring additional energy usage to overcome rolling resistance and air drag at that higher speed. In contrast, cutting the throttle immediately at 0 s hold allows the car to start coasting sooner, spending more time with the motor drawing no power. Second, there is an optimal range for the coasting minimum speed. Coasting down too little e.g. only to 5.0 m/s means the car doesn't exploit the full potential of coasting, requiring many short pulses and thus higher energy overall. Coasting down too much to 4.0 m/s extends the distance per pulse but also means fewer and more energy intensive pulses that may not fully compensate for the longer coasting. In our data, allowing the car to slow to about 4.5 m/s before re-accelerating struck the best balance, yielding the lowest energy per lap. At this 4.5 m/s threshold with no throttle hold, the vehicle would perform roughly 12–13 pulse cycles per lap, and the computed energy requirement of around 4939 J per lap is about 27% lower than the best 1 s hold case, and 50% lower than strategies with very long holds with approaching a constant speed approach. These findings validate that a well executed pulse and glide strategy can greatly reduce energy consumption compared to a constant speed run with the given motor.

4.3 Driving Strategy Result

The outcome of this evaluation directly informs the driving strategy that will be used in the Shell Eco-marathon competition in June. Based on the testing, the team will adopt the pulse and glide method with no throttle hold at the target speed and a reapply speed of around 4.5 m/s. This combination was

identified as the most energy efficient, minimising the energy consumed per 1319 m lap. In practice, the driver will accelerate the car to the desired top speed at around 6.5 to 7 m/s, then completely release the throttle and coast until the speed falls to 4.5 m/s before repeating the cycle. By minimising energy per lap in this manner, the vehicle can maximise its overall distance on the limited energy budget. In summary, although hardware upgrades were not implemented, the optimised driving strategy alone is expected to significantly improve the car's efficiency. These results will guide our race day approach, ensuring that we leverage the pulse driving technique to achieve the lowest possible energy consumption per lap and a competitive performance in the 2025 Shell Eco-marathon.

4.4 Driving Feedback Sub-System

To support real time adherence to the optimised driving strategy, a visual feedback system was developed using three indicator LEDs. This system receives live speed data from the onboard dynamometer and translates strategic speed thresholds calculated above into visual cues for the driver while driving.

The speed values are taken from the on board wheel sensor. These speed values, expressed in kilometres per hour, are transmitted to an Arduino Nano located in the cockpit. The Nano is responsible for interpreting the incoming speed data and illuminating one of three LEDs based on predefined thresholds derived from the driving strategy analysis.

The thresholds are as follows.

- Speed < 4.5 m/s: Red LED turns ON indicating the vehicle is under target speed and the driver may accelerate.
- $4.5 \leq \text{Speed} < 5.0$ m/s: Yellow LED turns ON indicating optimal speed range and the driver should maintain current input.
- Speed ≥ 6.8 m/s: Green LED turns ON indicating the car is above optimal speed and the driver should ease off the throttle.

This LED interface offers a lightweight, low power and distraction free method for the driver to stay within the efficiency envelope determined by simulation and data analysis. Unlike digital screens or complex feedback mechanisms, the simplicity of the traffic light inspired system ensures clarity during race conditions while aligning tightly with the strategic speed bands.

The system significantly improves the driver's ability to execute the planned acceleration and coasting profiles during a race lap while enabling more consistent energy management and helping to meet the vehicle's performance and efficiency goals.

5 Project Conclusion

5.1 Conclusion and Reflection

This project set out to investigate whether upgrading the GEEC's powertrain from a 24 V brushed DC motor to a 48 V system either brushed or brushless could lead to meaningful gains in energy efficiency and competition performance. Through a combination of simulation, dyno and track testing, component analysis, and strategy modelling, the project achieved a detailed evaluation of the car's electrical and mechanical systems, identifying several areas for improvement and future development.

Although time constraints prevented the full implementation of a new motor and controller within the 2024/25 cycle, the research and technical groundwork established during this project has laid a clear path forward. The selection of a high efficiency 48 V brushless motor, alongside a proposed custom built controller design, demonstrates the potential to dramatically reduce energy consumption and improve system responsiveness. Simulations and real world comparisons confirmed lower current draw, reduced ripple, and significantly improved torque to weight ratios giving this motor selection higher efficiency and a theoretical smaller overall energy consumption over a race which are critical advantages for an energy limited vehicle like the GEEC.

With hardware changes postponed, a shift in focus toward driving strategy resulted in another impactful development of the project. Through dyno and track based testing, a data driven driving approach was designed and validated. A pulse and glide strategy, with no throttle hold and a reapply speed of 4.5 m/s, emerged as the most efficient way to complete a race lap, by reducing energy consumption per lap by over 25% compared to constant speed driving. This driving profile will be adopted by the team at the 2025 Shell Eco-marathon in Poland and represents a major opportunity to improve competitive results without hardware upgrades.

In addition, this project further analysed the limitations of the current controller which was originally based on a non synchronous buck design and mapped out a more advanced BLDC controller architecture for future use. By considering component level losses, e.g. MOSFET efficiency, diode drop, and switching behaviour, and system level improvements such as regenerative braking, this research has contributed to the long term evolution of the GEEC platform.

Ultimately, this project has reaffirmed that marginal gains in efficiency whether through smarter electronics or better driving tactics can have a major impact in an ultra competitive environment like the Shell Eco-marathon. By combining detailed simulation, practical testing, and forward looking design work, the 2025 GEEC electrical subsystem is more understood, more capable, and more prepared for future success than ever before. The work carried out here will directly inform the team's

performance in this year's race and guide development for the next generation of students taking the car forward.

5.2 Recommendations For Future Work

While this project has delivered a detailed analysis of the GEEC's powertrain and driving strategy, several high impact developments remain for future implementation and testing. The following recommendations outline the most promising areas for further work and refinement.

5.2.1 Procurement and Integration of the 48 V Brushless Motor

The selected Maxon EC-i 52 brushless DC motor (634043) showed clear potential to improve torque output, reduce energy consumption and improve system voltage margins. Purchasing and physically integrating this motor into the GEEC drivetrain is the next critical step. Once installed, comprehensive validation through bench testing and track testing should be undertaken to compare actual performance with simulation predictions.

5.2.2 Custom Motor Controller Design for BLDC Operation

A custom built three phase motor controller tailored to the selected BLDC motor should be developed. This controller should include appropriate gate drivers, high efficiency MOSFETs and control logic to enable field oriented control (FOC), regenerative braking and precise torque management. Special attention should be given to PCB thermal design, protection circuitry, and deadtime control. Once operational, the controller should undergo full hardware testing, thermal profiling, and integration with the vehicle's new telemetry systems.

5.2.3 Reassessment of Driving Strategy with the New Powertrain

Once the new motor and controller are operational, the existing pulse and glide driving strategy should be revisited. Since the BLDC motor is expected to have a different efficiency map and dynamic response, the optimal pulse length, reapply speed, and throttle hold duration may change. New test runs should be conducted to recalibrate the driving strategy based on real data, ensuring that the full benefit of the improved powertrain is realised.

5.2.4 Development of a New Dynamometer Interface

The current dynamometer interface presents limitations in both update rate and measurement accuracy, particularly due to the slow and imprecise speed sensor currently in use. A revised dyno system should be developed to include a higher resolution speed sensing method e.g., optical encoder or magnetic ring with more poles and faster data acquisition. This will improve the quality of acceleration profiles and energy consumption readings, leading to more accurate driving strategy development and motor controller validation.

5.2.5 Enhanced Data Acquisition and Telemetry

Expanding the onboard DAQ system to include high frequency logging of motor current, voltage, throttle position, and vehicle speed will improve post race analysis and allow real time performance monitoring. Integration with the new motor controller via serial communication could also enable logging of internal controller parameters such as current, motor temperature estimates and fault conditions for live data displays.

By addressing these areas, future iterations of the GEEC can build on the current project's foundation and pushing towards even greater energy efficiency, system reliability and race day performance.

5.3 Social Impact

The Galway Energy Efficient Car project represents more than just a student engineering challenge, it is a platform for innovation that contributes meaningfully to the global effort to reduce transport related emissions and promote sustainable mobility. By pushing the boundaries of energy efficiency in electric vehicle design, this project serves as a proof of concept for what is achievable when intelligent control systems, lightweight engineering, and data driven strategy come together. Every incremental improvement in the GEEC's efficiency directly contributes to the broader understanding of how EVs can be made more effective, practical and environmentally responsible.

In the context of the climate crisis, developing ultra efficient electric vehicles is not just a technical achievement but it is a need to society. Transport accounts for a significant portion of global greenhouse gas emissions and electric vehicles offer a clear pathway toward reducing this impact. However, making EVs more efficient is essential for extending battery life, reducing material demands and lowering charging costs, particularly in regions where renewable energy is not yet the dominant source of electricity. The GEEC project demonstrates how efficiency gains whether through improved driving strategy, powertrain optimisation, or system-level design can reduce energy usage per meter and ultimately decrease the environmental footprint of electric mobility. Beyond its technical goals, the GEEC acts as a public facing initiative to promote awareness around sustainable transport. Through open days, media coverage and outreach events, the car is showcased as an example of student led innovation with real world relevance.

In summary, the GEEC is not just about building a better car it's about challenging assumptions, testing ideas and proving that sustainability and performance can coexist. The work carried out in this project helps inform the future of transport and contributes to a more energy resilient and environmentally sustainable society. Through continued innovation and student engagement the GEEC has the potential to leave a lasting social and ecological impact far beyond the competition circuit.

5.4 Project Management

Effective project management was essential to the success of this year's GEEC work, given the collaborative nature of the team and all the individual and sub systems within the vehicle. To ensure steady progress, a structured meeting schedule was maintained throughout the year. This included weekly full team meetings to discuss overall progress, address integration issues and ensure alignment across all technical areas. These meetings were particularly important for coordinating work between subsystems such as power electronics, control, and mechanical teams. Unfortunately I was unable to attend as frequently as I should have but always caught up with the other project leaders.

In addition to the main team meetings, we held weekly supervisor meetings to report on individual and team progress, receive guidance and adjust priorities as needed. A smaller group of core contributors also met separately in weekly student meetings allowing for focused technical discussions and direction the students want the projects to go.

To support planning and accountability, we used a centralised project dashboard that tracked key milestones, deadlines and checklists across the development timeline. This visual tool made it easy to assign responsibilities, monitor ongoing tasks, and identify delays before they impacted the schedule. It was especially useful during integration phases and helped ensure that critical tests and simulations were completed on time.

As a final year team member with leadership responsibilities, I placed strong emphasis on task delegation. By involving other team members in hands-on work such as soldering, sensor calibration, and test rig assembly. I was able to focus on system level design and strategy development, while also helping build the experience and confidence of future GEEC contributors.

In summary, the project benefited significantly from structured communication, a shared planning tool, and a clear division of work. These management practices ensured that technical progress was steady and that the team remained aligned throughout the development cycle.

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7 Appendix

7.1 Dyno Code

```
#include "HX711.h"
```

```
#include <SoftwareSerial.h>
```

```
// Hall effect & load cell pins
```

```
const int hallEffectPin = 18; // interrupt pin
```

```
const int CLK = 2;
```

```

const int DOUT = 3;

// HX711 load cell object

HX711 loadcell;

float calibration_factor = -143.56; // Adjust based on calibration

float torqueDistance = 0.20;    // m, from load cell to dyno wheel center

// SoftwareSerial to Nano (D11 = TX)

SoftwareSerial nanoSerial(10, 11); // RX, TX (TX → Nano D4)

// Timing

int sampleTime = 1000; // ms between samples

long startTime;

long count = 1;

long prevTime;

// RPM calculation

int wRPMcount = 0;

double wTime_current = 0;

double wTime_old = 0;

double rpm = 0;

int numMagnets = 20;    // magnets on wheel

double radius = 0.251; // m, wheel radius

```

```

void setup() {

    loadcell.begin(DOUT, CLK);

    loadcell.set_scale(calibration_factor);

    loadcell.tare();


    Serial.begin(9600);    // To PC

    nanoSerial.begin(9600); // To Nano


    attachInterrupt(digitalPinToInterrupt(hallEffectPin), wRPMDetectorInterrupt, RISING);


    startTime = millis();

    prevTime = startTime;


    Serial.println("count,time(ms),rpm,speed(m/s),torque(Nm),power(W)");

}


void loop() {

    if (millis() - prevTime >= sampleTime) {

        // Read data

        double load = loadcell.get_units(10); // grams

        double rpm = calcRPM();           // revolutions per minute

        double speed = msConvert(rpm);    // m/s
    }
}

```

```

double torque = torqueDistance * (load / 1000.0) * 9.81; // Nm

double power = calcMechPower(rpm, torque);           // Watts

long time = millis() - startTime;

// Output to Serial Monitor

Serial.println(String(count) + "," + String(time) + "," + String(rpm) + "," + String(speed) + "," +
String(torque) + "," + String(power));

// Send speed to Nano

nanoSerial.println(speed);

Serial.println("Sent speed to Nano: " + String(speed));

count++;

prevTime = millis();

}

}

void wRPMDetectorInterrupt() {

    wRPMcount++;

}

double calcRPM() {

    detachInterrupt(digitalPinToInterrupt(hallEffectPin));

```

```

wTime_current = (millis() / 1000.0);

rpm = (60.0 * (wRPMcount / (wTime_current - wTime_old))) / numMagnets;

wTime_old = wTime_current;

wRPMcount = 0;

attachInterrupt(digitalPinToInterrupt(hallEffectPin), wRPMDetectorInterrupt, RISING);

return rpm;

}

double msConvert(long RPM) {

    return (RPM * ((2 * 3.14159) / 60.0) * radius);

}

double calcMechPower(double rpm, double torque) {

    double radSSec = 2 * PI * rpm / 60.0;

    return radSSec * torque;

}

```

7.2 Throttle Code

```

#include <SoftwareSerial.h>

// Serial input from Mega: RX = D4, TX = D5 (TX not used here)

SoftwareSerial speedSerial(4, 5);

// Configurable throttle timing (in seconds)

float acclTime = 5;

```

```

float steadyTime = 25;

float decclTime = 5;


// Computed timing values

int acclSubTime = acclTime * 10;    // 0–100 ramp = 100 steps

int steadyIntTime = steadyTime * 1000;

int decclSubTime = decclTime * 10;


const int throttlePin = 3;

const int startButton = 2;

boolean buttonPressed = false;


// LEDs

const int led1 = 5; // Speed < 4.5

const int led2 = 6; // 4.5 ≤ Speed < 5.0

const int led3 = 7; // Speed ≥ 6.8


void setup() {

    attachInterrupt(digitalPinToInterrupt(startButton), buttonISR, FALLING);


    Serial.begin(9600);    // Debug monitor

    speedSerial.begin(9600); // Receiving from Mega


    pinMode(led1, OUTPUT);

    pinMode(led2, OUTPUT);

    pinMode(led3, OUTPUT);

```

```
pinMode(throttlePin, OUTPUT);
```

```
Serial.println("Throttle controller ready...");
```

```
}
```

```
void loop() {
```

```
    // Handle incoming speed data
```

```
    if (speedSerial.available()) {
```

```
        float speed = speedSerial.parseFloat();
```

```
        if (speed > 0.01) {
```

```
            Serial.print("Received speed: ");
```

```
            Serial.println(speed);
```

```
            updateLEDs(speed);
```

```
        }
```

```
    }
```

```
    // Run throttle routine
```

```
    if (buttonPressed) {
```

```
        Serial.println("Starting throttle run...");
```

```
        // Acceleration ramp
```

```
        for (int i = 0; i < 100; i++) {
```

```
            analogWrite(throttlePin, 55 + 2 * i); // Smooth ramp
```

```
            delay(acclSubTime);
```

```
        }
```



```

// Steady state

analogWrite(throttlePin, 255);

Serial.println("Holding steady...");

delay(steadyIntTime);


// Deceleration ramp

for (int i = 0; i < 100; i++) {

    analogWrite(throttlePin, 255 - 2 * i);

    delay(decclSubTime);

}

analogWrite(throttlePin, 0);

Serial.println("Throttle run complete.");

buttonPressed = false;

}

}


// ISR for button press

void buttonISR() {

    buttonPressed = true;

}


// LED status based on speed thresholds

void updateLEDs(float speed) {

    digitalWrite(led1, LOW);

```

```
digitalWrite(led2, LOW);  
  
digitalWrite(led3, LOW);  
  
if (speed < 4.5) {  
    digitalWrite(led1, HIGH);  
    Serial.println("LED1 ON (speed < 4.5)");  
} else if (speed < 5.0) {  
    digitalWrite(led2, HIGH);  
    Serial.println("LED2 ON ( $4.5 \leq \text{speed} < 5.0$ )");  
} else if (speed  $\geq 6.8$ ) {  
    digitalWrite(led3, HIGH);  
    Serial.println("LED3 ON ( $\text{speed} \geq 6.8$ )");  
} else {  
    Serial.println("Speed in target range, no LED active.");  
}  
}
```